

Evaluating Land Use and Local Temperature Dynamics of Dutch Solar Parks Using Remote Sensing

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Abstract

The expansion of solar parks in the Netherlands as part of the energy transition raises questions about spatial planning in a densely used landscape. Through land use and land cover change, solar parks interact with broader landscape development and alter local surface energy balances. The local impacts of solar parks on land surface temperature (LST) remain understudied in the Dutch context, a gap that can be addressed using remote sensing. This study examines land-use change and landscape stability around Dutch solar parks and quantifies associated variations in local LST to better understand how solar parks shape their surrounding environment. A geodatabase of 206 Dutch solar parks ≥ 4 ha was derived from existing datasets and enriched with key characteristics to support analysis of land-use and LST changes. Land-use change associated with solar parks is assessed using national land-use datasets prior to and post-solar park development, examining changes within and around each park to quantify local and broader landscape stability. Land surface temperature (LST) effects are evaluated using Landsat 8 thermal imagery, comparing temperature changes at the parks with surrounding control areas to detect temperature change attributable to the installations. These temperature effects are related to prior land use of the park sites. Finally, the relationship between solar park characteristics, land use change and temperature change is explored through a multiple linear regression (MLR). Results show that most solar parks were developed on agricultural land, contradicting national policy. Average landscape stability adjacent to (500 m around) the parks and in the broader landscape (500 m – 2.5 km) around the parks is consistently lower than the national landscape stability, but there is large provincial variety. The LST analysis shows a mean additional warming in solar park areas of 0.69°C , but there is a large variation in this additional temperature change, with 77 parks showing relative cooling. Results of the MLR indicate that solar parks on water consistently exhibit higher additional warming, while parks located within urban-dominated 500 m buffers show slight cooling. Overall, the model explains only a modest portion of $\Delta\Delta T$ variability, highlighting that local thermal responses are highly context dependent. These findings suggest that solar parks primarily integrate into already dynamic landscapes rather than highly stable areas, and that prior land use and immediate landscape context play a key role in local thermal impacts. Implications include potential effects on microclimate, soil processes, vegetation growth, and habitat quality, particularly in sensitive areas. Future research could examine technical design specifics, park management practices, and day/night temperature variation, and expand the analysis to include smaller parks and rooftop PV to guide climate-sensitive and ecologically informed solar park planning in the Netherlands.

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1. Introduction

1.1 Climate and energy transitions

The latest Climate and Energy Outlook (Klimaat- en Energieverkenning), which tracks the Netherlands' progress on climate policy each year, shows that the government is failing to take the necessary measures to phase out CO₂ emissions, or even to reduce them in line with its own targets (PBL, 2025). The planned 55% reduction by 2030 (compared to 1990 levels) has a less than 5% likelihood of being achieved. Meanwhile, the climate and biodiversity crises continue to escalate, with their impacts already clearly visible: average temperatures and sea levels are rising, extreme weather events such as droughts and heavy rainfall are becoming more frequent, and ecosystems are collapsing as flora and fauna die out or face extinction (IPCC, 2023).

Renewable energy plays a crucial role in mitigating the climate crisis, but, in this sector too, the Netherlands is projected to miss its next target: a 39% share of renewables in the national energy mix by 2030 (European Commission, 2023; PBL, 2025). This target represents the Dutch contribution to the binding EU-wide goal of achieving 42.5% renewable energy by 2030 (EU, 2023). Nevertheless, the share of renewables is expected to increase substantially even under the current policy framework (PBL, 2025) (Figure 1). This shift is crucial not only for reducing CO₂ emissions from fossil fuel combustion but also for strengthening Dutch energy independence, as nearly 80% of current energy consumption relies on imports (PBL, 2025).

Renewable electricity generation in the Netherlands, 2015-2023, and forecast 2024-2030

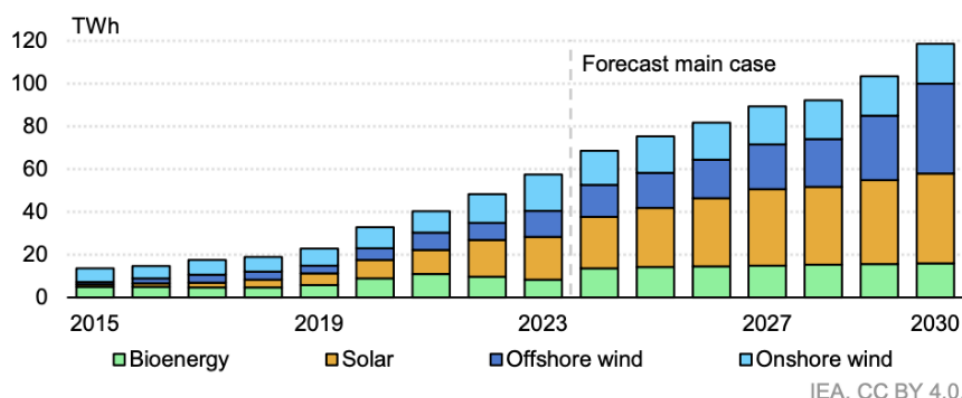


Figure 1: Renewable electricity generation in the Netherlands, 2015-2023, and forecast 2024-2030 (source: IEA (2024-a), Renewable Energy Progress Tracker, CC by 4.0)

Looking beyond 2030, the Dutch government has begun to sketch the contours of the long-term energy transition. The 2023 National Energy System Plan (Nationaal Plan Energiesysteem) sets out a vision for the system up to 2050. It prioritizes maximizing renewable energy supply, supported by state funding, on the premise that demand will adapt to the growth of renewable generation and infrastructure (Rijksoverheid, 2023). Moreover, the 2024 Energy Infrastructure Programme (Programma Energiehoofdstructuur) outlines where the new energy infrastructure towards 2050 should be located on the pathway to a climate-neutral energy system by 2050 (Rijksoverheid, 2024).

1.2 The rise of solar photovoltaics in the Netherlands

In the past decade, Solar Photovoltaics (PV) have emerged worldwide as the fastest-growing renewable technology, with global capacity tripling between 2018 and 2023. It is expected to overtake all other sources to become the largest contributor to renewable electricity generation before 2030 (IEA, 2024-b). While solar power may not seem an obvious fit for the Netherlands, the country is at the forefront of this trend. By 2030, solar PV is projected to account for nearly three-quarters of new renewable capacity additions (IEA, 2025). Remarkably, the Netherlands even had the largest share of solar PV in its electricity capacity in the world in 2022 (36%), coming way up from its modest position just five years earlier in 2017 (Figure 2) (IEA, 2025). By 2024, installed solar PV capacity in the Netherlands reached 28.6 gigawatt-peak (GWp), with businesses responsible for nearly 60% of installations (CBS, 2025) (Figure 3). Moving forward, the government has set the goal of generating at least 35 terawatt-hours (TWh) of renewable electricity by 2030 from large-scale solar projects (>15 kilowatt-peak, kWp) and onshore wind, alongside an additional 7 TWh from smaller solar systems (<15 kWp) in the built environment (RVO, 2025).

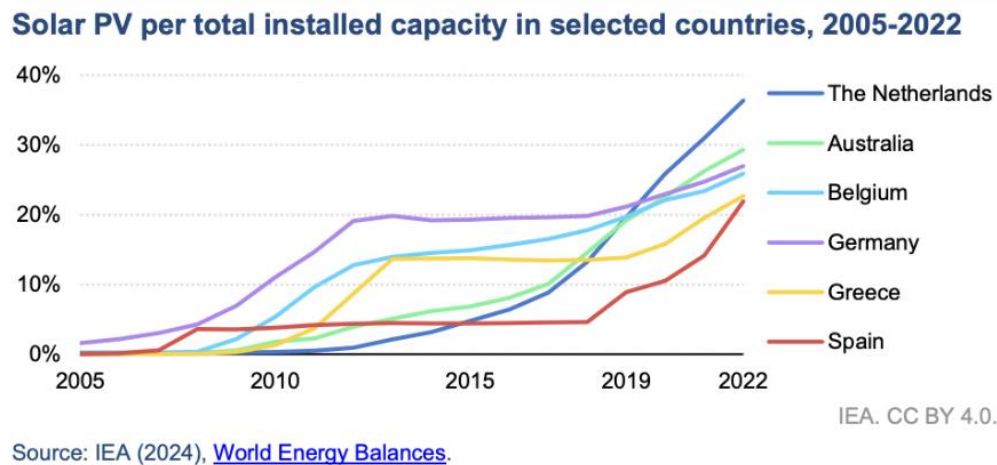


Figure 2: Solar PV per total installed capacity in The Netherlands, Australia, Belgium, Germany, Greece and Spain, 2005-2022 (source: IEA (2024), *World Energy Balances*, CC by 4.0)

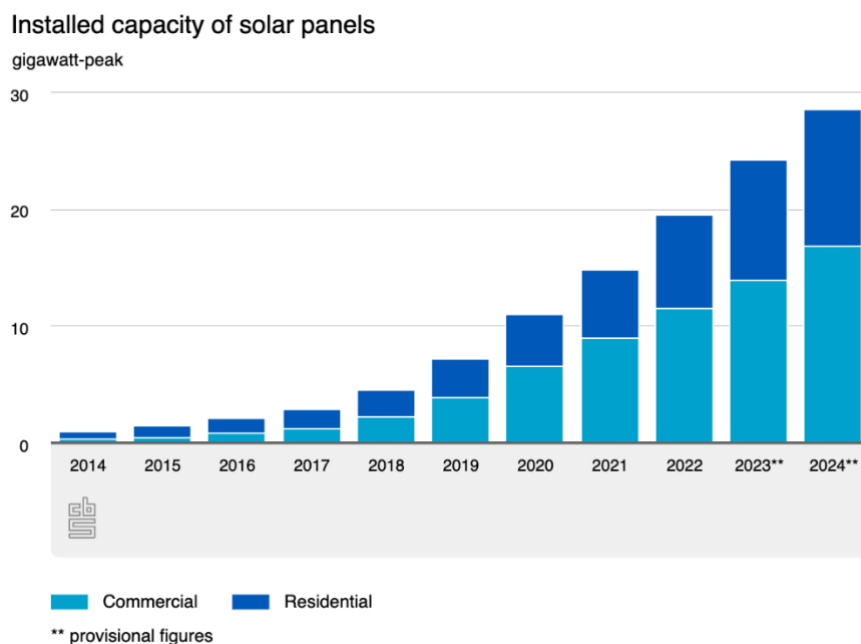


Figure 3: Installed capacity of solar panels in the Netherlands (in GWp), commercial vs. residential (source: CBS, 2025, CC by 4.0).

However, the rapid growth of solar PV has intensified debates over spatial planning. The largely market-driven development of solar parks, combined with limited spatial guidance from governments, has led to growing concerns about the appropriate siting of installations in a country where space is already scarce (de Jonge, 2023; RVO, 2025). In response, the Dutch government introduced the so-called “solar ladder” (*zonneladder*) in the 2020 National Spatial

Strategy (Nationale Omgevingsvisie), later reinforced through additional agreements in 2023. This policy framework sets a clear hierarchy for solar panel installation: rooftops first (step 1), followed by sites in and just outside built-up areas (steps 2 and 3), and only then rural locations, with a strong preference for multifunctional land use (step 4). The approach explicitly aims to preserve farmland and nature as much as possible. Under current these mentioned spatial regulations, the use of agricultural or natural land is in principle undesirable and only permitted in exceptional cases, such as agri-PV systems that combine energy generation with continued farming, designated transition areas, or projects that demonstrably contribute to reducing grid congestion (RVO, 2025). Within this policy context, ground-mounted solar parks have nevertheless expanded rapidly, becoming an increasingly visible component of the Dutch energy landscape.

2. Theoretical Framework

2.1 Landscape stability around solar parks

In a more general sense, the expansion of ground-mounted solar parks invites closer examination of how renewable energy infrastructure interacts with ongoing landscape development in the Netherlands. Solar parks are not developed in isolation; they are embedded within landscapes that are already shaped by long-term processes of agricultural restructuring, urban expansion, nature restoration, and infrastructure development. Understanding the implications of solar park deployment therefore requires attention not only to the park footprint itself but also to the surrounding landscape context in which it is introduced.

The concept of landscape stability can be used to describe the extent to which land-use patterns around solar parks remain relatively persistent over time or are subject to change. The introduction of large-scale renewable energy infrastructure into stable agricultural or natural landscapes may present a more profound intervention than similar developments in landscapes that are already undergoing functional or spatial change. Examining landscape stability allows this to be explored empirically by assessing whether solar parks tend to be located in relatively stable landscapes or in areas already characterized by land-use transitions and multifunctionality.

Landscape stability also forms an important contextual layer for assessing other environmental effects of solar parks. Land-use and land-cover composition strongly influence surface characteristics such as vegetation cover and soil moisture, both of which play a role in regulating local climate processes. Changes associated with solar park development therefore occur within a landscape matrix that may either amplify or dampen environmental impacts.

2.2 The interaction of solar PV with its environment

At the site level, the development of solar parks inevitably leads to land use and land cover changes on the solar park sites, which in turn can alter the local surface energy balance and meteorological conditions (Heusinger et al., 2020). Changes in the surface energy balance directly affect local climate conditions, including (near-)surface air temperatures, soil heating, and evapotranspiration. Figure 4 presents a schematic visualization of the surface energy balance. These effects can influence both environmental and societal outcomes: They may

affect local ecosystem functioning, local climate comfort, and the effectiveness of land management strategies. In the context of solar park development, understanding how energy fluxes are redistributed is essential to evaluate the environmental consequences of this form of renewable energy expansion.

When PV panels are installed, a portion of the incoming solar (shortwave) radiation is converted into electricity, effectively removing this energy from the surface-atmosphere system. Simultaneously, some of the absorbed solar energy is converted into heat within the PV modules. While the extraction of energy for electricity generation might be expected to produce a local cooling effect—by reducing the energy available to heat the surface and lower atmosphere—the increased sensible heat flux (increase in temperature) from the warmed PV modules can offset this cooling and lead to local warming. The magnitude of these effects depends on PV efficiency and operating temperature (Heusinger et al., 2020).

The remainder of incoming solar radiation is reflected back to the atmosphere, with the proportion of reflection determined by the panels' albedo. Since PV panels are typically darker than natural surfaces, their albedo is lower, and thus they absorb more radiation than many natural land covers (Heusinger et al., 2020).

PV panels also shade the ground below them, reducing the amount of solar radiation reaching the surface. As a result, ground surface temperatures under panels can be lower, and thermal (longwave) radiation emitted from the shaded ground is reduced. This shading decreases the ground's sensible heat flux; however, as noted, heating of the PV panels increases their own sensible heat flux, meaning that solar PV installations can locally enhance near-surface air temperatures even while shading the surface (Heusinger et al., 2020). The extent of shading and its effects depend strongly on panel spacing, tilt and tracking (the ability to move towards the sun) (Heusinger et al., 2020).

PV installations further modify thermal radiation patterns. Panels emit thermal radiation upward and downward, exchange radiation with adjacent panels in arrays, and generally have slightly lower emissivity than natural surfaces (Xu et al., 2024). Consequently, PV arrays retain heat more efficiently, especially at night, which can contribute to higher local temperatures. Figure 5 illustrates the PV energy balance schematically.

Changes in vegetation cover (decreasing) under and around the panels can also reduce soil and plant evaporation (latent heat), further affecting the surface energy balance and local cooling potential (Guoqing et al., 2021). In addition, local airflow patterns may be altered by the presence of PV arrays, influencing the horizontal distribution of heat and energy, potentially beyond the solar park boundaries (Guoqing et al., 2021).

Overall, the question of whether solar PV systems produce local cooling or warming does not have a straightforward answer. The net effect depends, among other things, on the

specific design of the PV park, including panel arrangement, land cover, and operational characteristics.

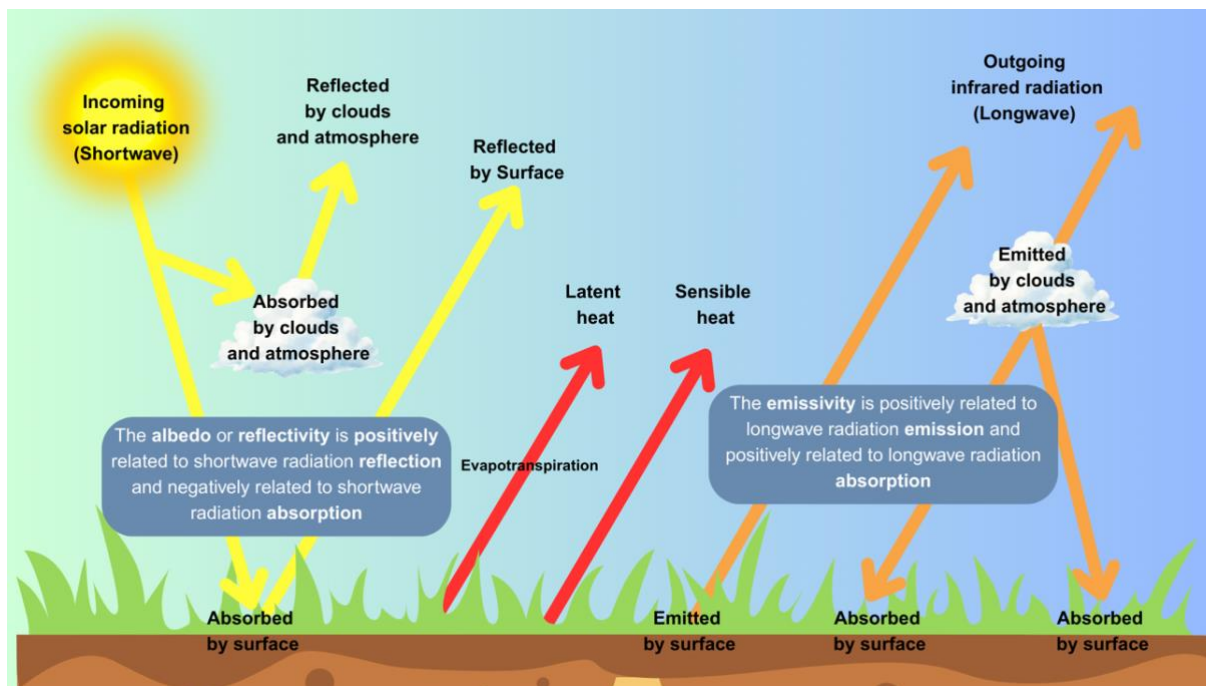


Figure 4: Surface energy balance (source: created by the author)

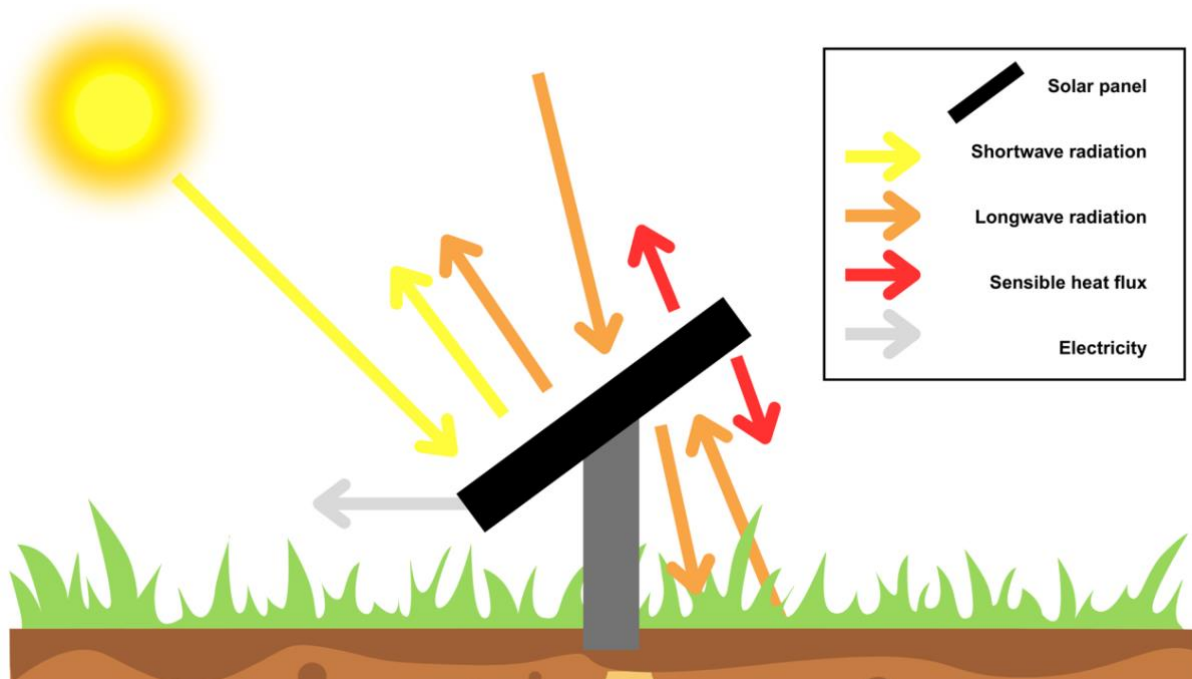


Figure 5: PV energy balance (source: created by the author)

2.3 Assessing local environmental effects using remote sensing

Responding to parliamentary inquiries about the rise of solar park development in the Netherlands and their potential effects on agriculture, soil, biodiversity, and landscape quality, Wageningen Environmental Research examined 25 solar parks between 2019 and 2020 (Schotman et al., 2021). The study found wide variation in vegetation and biodiversity outcomes, depending largely on management practices. A more recent study by Becker et al. (2025) assessed 46 solar parks in Groningen, focusing on land use competition, biodiversity loss, and landscape perception. They found that most solar parks were developed on former agricultural land (59%)—raising concerns about alignment with the “solar ladder” policy.

However, while studies set in the Netherlands have examined some environmental aspects of solar park development—focusing mainly on soil quality, biodiversity, and landscape values (e.g., Schotman et al., 2021; Treasure et al., 2025)—most research to date has concentrated on sociopolitical dimensions such as community acceptance (van den Berg & Tempels, 2022), the social feasibility of floating PV (Bax et al., 2022), and impacts on housing values (Dröes & Koster, 2021). Thus, the local biophysical impacts of solar parks, particularly their place within broader landscape development and their influence on the local energy balance, remain understudied in the Dutch context.

Remote sensing offers a practical approach that can address this gap. It allows researchers to track land surface conditions across space and time, providing consistent measurements for multiple sites simultaneously. In particular, Land Surface Temperature (LST) products derived from MODIS, Landsat, and other sensors make it possible to quantify how solar park development alters local thermal environments. International studies have investigated these impacts outside of the Netherlands (Xu et al. 2024; Zhang & Xu, 2020).

Xu et al. (2024) used MODIS Land Surface Temperature (LST) products from Aqua and Terra (MYD11A2 and MOD11A2) at 1-km spatial resolution and 8-day temporal resolution, covering four daily overpass times since 2002, to assess the local effects of 116 solar farms across the world, but not in higher latitudes similar to the Netherlands, with a minimum area of 400 ha or 4 km². They compared the solar farm measurements to buffer areas that they generated as 1-km-wide ring-shaped zones around the solar farm, 1 km away from the solar farm boundaries, based on evidence that solar farm impacts extend less than 1 km beyond the installation (Guoqing et al., 2021).

Similarly, Zhang & Xu (2020) analyzed 23 large PV powerplants in China (latitudes 9–43°, areas 5.4–91.9 km²) using MODIS LST data with 2-km control area buffers to account for background climate trends. They calculated the effect of PV powerplants on surface

temperature by comparing differences before and after construction relative to the control area.

Both studies found significant land surface cooling, especially during the daytime, but LST impacts were found to be very dependent on prior land use (Xu et al. 2024). Barren land experienced the strongest cooling, followed by grassland, while cropland showed the weakest cooling and, in some cases, nighttime warming (10 of 22 cropland sites). Zhang & Xu (2020) also found some warming effects during the night, specifically in dry and cold environments. Both studies also found that daytime cooling decreases with latitude (though in Zhang & Xu's (2020) study only in the daytime); higher latitudes show weak cooling and little seasonality (Xu et al. 2024; Zhang & Xu, 2020). Notably, the Netherlands is located at an even higher latitude. Only 15/116 samples in Xu et al's (2024) study were in the highest latitudes (>40) (2 cropland, 3 barren, 10 grassland).

The cooling effects were primarily attributed to photovoltaic conversion efficiency and faster nighttime cooling due to higher thermal conductivity of PV panels (Xu et al., 2024). Additional contributing factors included reduced surface albedo, enhanced effective albedo, and shading from panels (Zhang & Xu, 2020). Furthermore, solar farm size was positively correlated with the magnitude of cooling (Xu et al., 2024; Zhang & Xu, 2020).

However, both studies highlighted limitations related to MODIS emissivity. The algorithm assumes fixed emissivity values for 18 landcover types (vegetation >0.9), whereas solar panels have emissivity around 0.8, leading to potential underestimation of LST over PV sites (Xu et al., 2024).

Zhang & Xu (2020) also addressed this emissivity bias, stressing the sensitivity of the satellite-measured surface temperature to surface emissivity. Most PV panels are glass-coated, with emissivity slightly lower than local soils. However, Zhang & Xu also argue that the cavity effect, which occurs when heat is trapped and re-emitted between panels, increases the collective emissivity of a solar park. As a result, according to the study, the overall emissivity of PV power plants closely matches that of the surrounding soil, especially in sandy or dry vegetated areas typical of PV installations. Therefore, they conclude that PV construction is unlikely to significantly alter surface emissivity. This is a small but significant contradiction to the claim made by Xu et al. (2024).

Across all studies, impacts were consistently shown to be highly dependent on local conditions, including climate, site characteristics, and management practices. Authors emphasize the need for further research into environmental effects and for the development of robust (including remote sensing-based) methodologies to assess the impacts of solar farms on local ecosystems.

2.4 General objective and research questions

The general objective of this thesis is to assess land-use change and landscape stability around Dutch solar parks, alongside variations in local land surface temperature.

To achieve this objective, the research focuses on the following questions (RQs):

1. What land-use changes are associated with solar park development, and how do these changes relate to immediate and broader landscape stability and policy goals?
2. How do solar parks influence local land surface temperatures?
3. To what extent are variations in land surface temperature within Dutch solar parks associated with prior land use and surrounding landscape stability?

3. Data

This study draws on multiple spatial datasets to analyze land-use change, landscape stability, and local land surface temperature (LST) around Dutch solar parks. Table 1 summarizes the spatial datasets used, including their sources, temporal and spatial resolutions, key attributes, and the purpose of each dataset in the context of this research. In addition to these datasets, technical project information (e.g., capacity, number of panels) was collected from solar park developers' websites to enrich the Top10NL base polygons. All solar park names were sourced from Wiki-Solar (Wiki-Solar, 2025). A full list of consulted websites for the additional solar park attributes is provided in Appendix A.

Table 1: *Spatial datasets sourced for this research*

Dataset	Source	Temporal Resolution	Spatial Resolution	Attributes	Purpose
Top10NL	Kadaster	Annual	Polygon	Functional areas	Base layer for solar parks
LGN Land Use Database (LGN7 and LGN2024 versions)	Wageningen Environmental Research	Annual	25 m	Land use classes	Land use change analysis
Landsat 8 Collection 2 Level-2 LST	USGS / NASA	16-day	100 m	LST	Temperature analysis
Administrative boundaries	CBS	Static	Polygon	National / provincial limits	Spatial joins / regional stats

4. Study area

4.1 Solar park database

A key prerequisite for this study is an up-to-date map of operational solar parks. The Top10NL dataset is part of the functional-area register (Basisregistratie Topografie, BRT) maintained by the Dutch Cadaster, Land Registry, and national mapping agency (Kadaster). Since 2019, this dataset has recorded solar parks as functional areas, though it typically lags behind in reality by at least a year due to rapid solar park expansion.

A geodatabase of Dutch solar parks was compiled by integrating multiple sources. Top10NL provided the base polygons, filtered to extract solar parks into a dedicated layer. To enrich the dataset, project-specific information, including floating or ground-mounted status, client, contractor, construction or commissioning year, installed capacity (MWp), and number of panels (where available), was added to enrich the Top10NL base polygons.

Only solar parks of ≥ 4 hectares (40,000 m²) were included to reduce mixed-pixel effects given Landsat's 100 m thermal resolution (relevant for RQ2), resulting in 246 parks. A spatial join with provincial boundaries allowed for the generation of regional statistics.

Finally, data preprocessing led to a reduction of the initial 246 solar parks > 4 ha to 206 parks due to some solar parks having no public information available and other parks being represented by two polygons, in which case they were merged.

4.2 Study area

Figure 6 shows the study area for this research, including the 206 solar parks registered in the database. The largest solar parks are located in Groningen, Flevoland, and Noord-Brabant, with areas of approximately 930,000 m², 830,000 m², and 788,000 m², respectively. Beyond these few large sites, park size decreases rapidly, with most solar parks larger than four hectares being smaller than 10 hectares (109 out of 206). These park area sizes should be interpreted cautiously, as they are based on the Top10NL polygon geometries, which do not always perfectly match the actual park footprints. Nevertheless, Figure 6 provides an overview of relative park sizes.

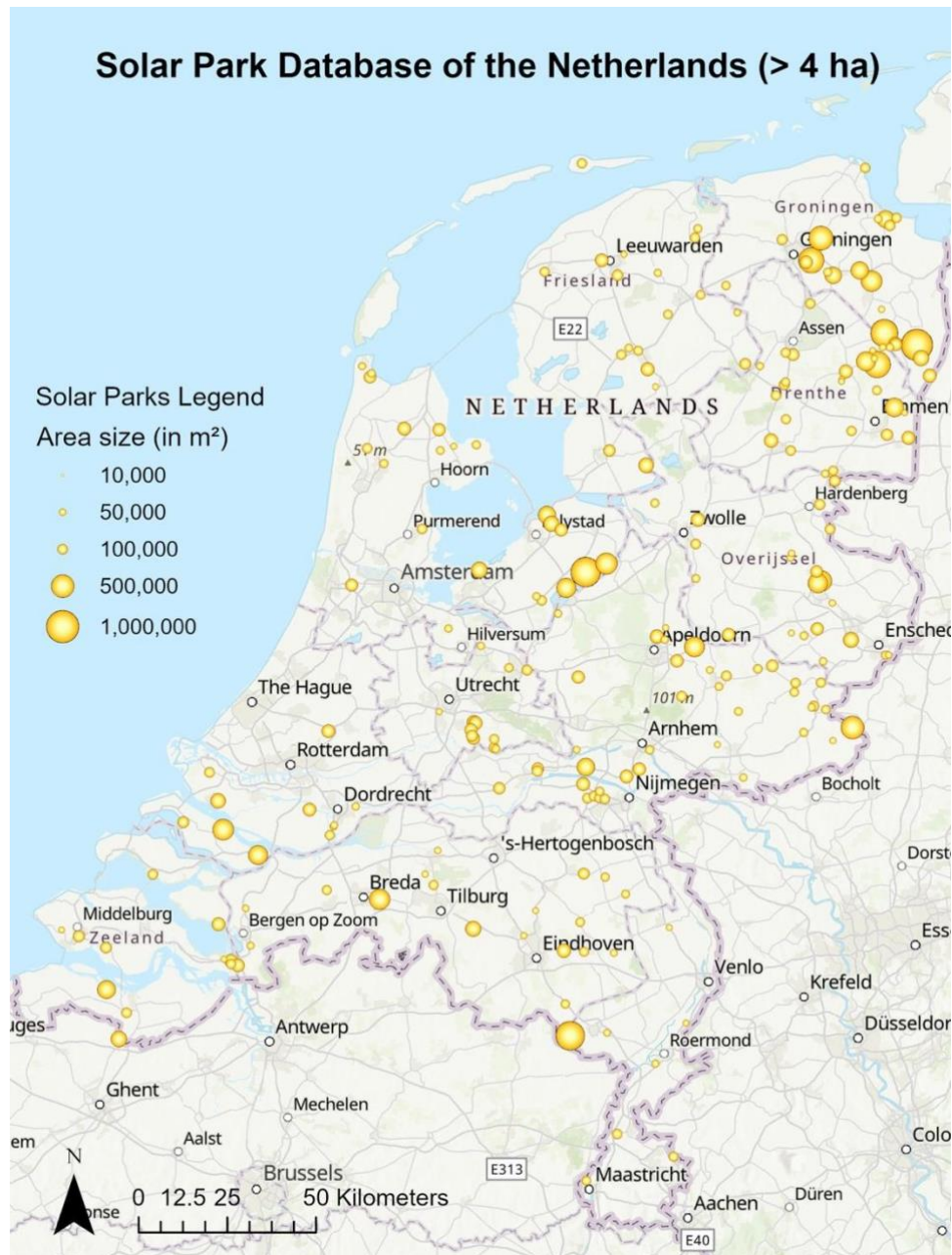


Figure 6: Study area in the Netherlands showing the 206 registered solar parks larger than 40,000 m², based on TOP10NL functional area data.

Table 2 and Figure 7a show the solar park count per province. Gelderland contains the highest number of solar parks larger than four hectares, followed by Drenthe and Groningen. This distribution accords with the fact that the northeastern Netherlands covers a larger area and is less densely populated than the southern and western regions.

Table 2: Solar park count per province

Province	Count
Gelderland	39

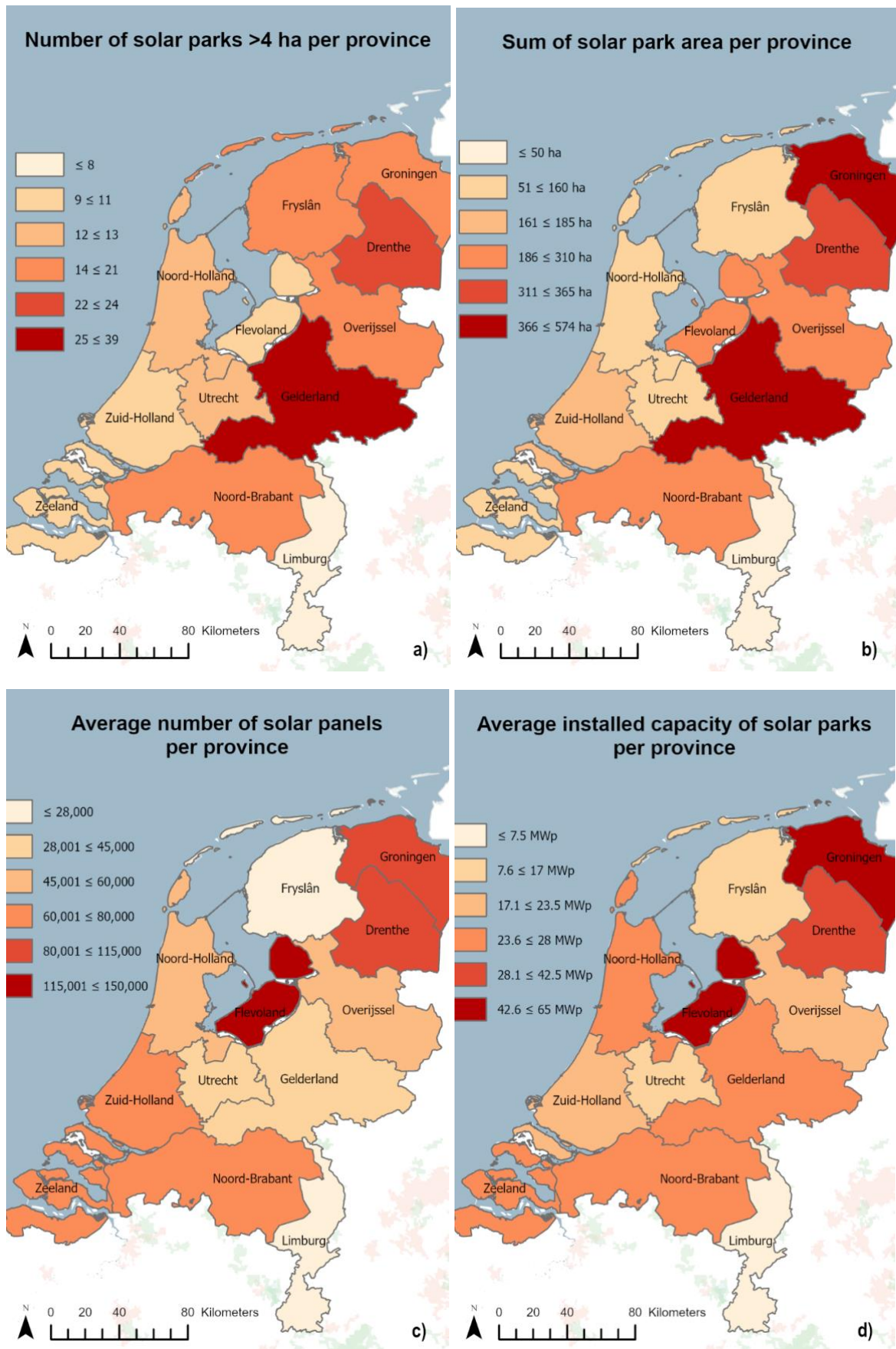
Drenthe	24
Groningen	21
Noord-Brabant	20
Overijssel	20
Fryslân	17
Noord-Holland	13
Utrecht	12
Flevoland	11
Zeeland	11
Zuid-Holland	11
Limburg	7
Total	206

In solar park area size, Groningen is the frontrunner, housing more than 570 hectares of large solar parks (> 4 ha), followed by Gelderland with more than 470 ha of solar parks and Drenthe with about 360 ha of solar parks (see Figure 7b). Thus, Groningen has some large parks compared to the other provinces.

The earliest years of construction and completion in the database are 2016. The latest year of construction is 2024 and the latest year of completion is 2025. The mean and median year of construction is 2020 and the mean and median year of completion is 2021.

The average number of installed panels is 44,415 (median 29,088) (no data = 31), with the max 312,148 and the min 45 panels (Figure 7c). The average installed capacity in megawatt peak (MWp) is 19.56 (median 12.5) with a maximum capacity of 144.2 MWp in Flevoland) and a minimum capacity of 2 MWp (in Gelderland) (Figure 7d). This example explains the slightly different spatial distribution of solar park area size versus average solar park capacity and number of panels.

The database contains 10 floating solar parks. Figure 8 shows the spatial distribution of these parks. The database contains floating solar parks (> 4 ha) in Overijssel (3), Drenthe (2), Fryslân (2), Gelderland (1), Groningen (1), and Limburg (1).





(Note: symbol sizes not directly comparable to Figure 6)

5. Methods

Figure 9 summarizes the overall workflow of this study, from the creation of the solar park geodatabase to the analyses conducted for each research question. For RQ1, spatial analyses assess land use and landscape stability using multiple buffers and derive metrics based on historical and recent land-use data. RQ2 focuses on changes in land surface temperature, with preprocessing and statistical analyses performed on satellite imagery to quantify park-specific temperature effects. RQ3 integrates outputs from the first two questions, exploring relationships between prior land use, landscape persistence, and temperature changes through statistical modeling. The flowchart highlights the sequence of data preparation, analytical steps, and the resulting outputs.

Spatial analyses for research question 1 were conducted in ArcGIS Pro (v3.5), while analyses for research question 2 were performed using Google Earth Engine. Data preprocessing, including the organization and cleaning of CSV files, as well as basic statistical calculations and table generation, was performed in Microsoft Excel. More complex statistical analyses were carried out in R (v4.5). A flowchart summarizing the research workflow, including data sources, preprocessing steps, and analytical procedures, is shown in Figure 9.

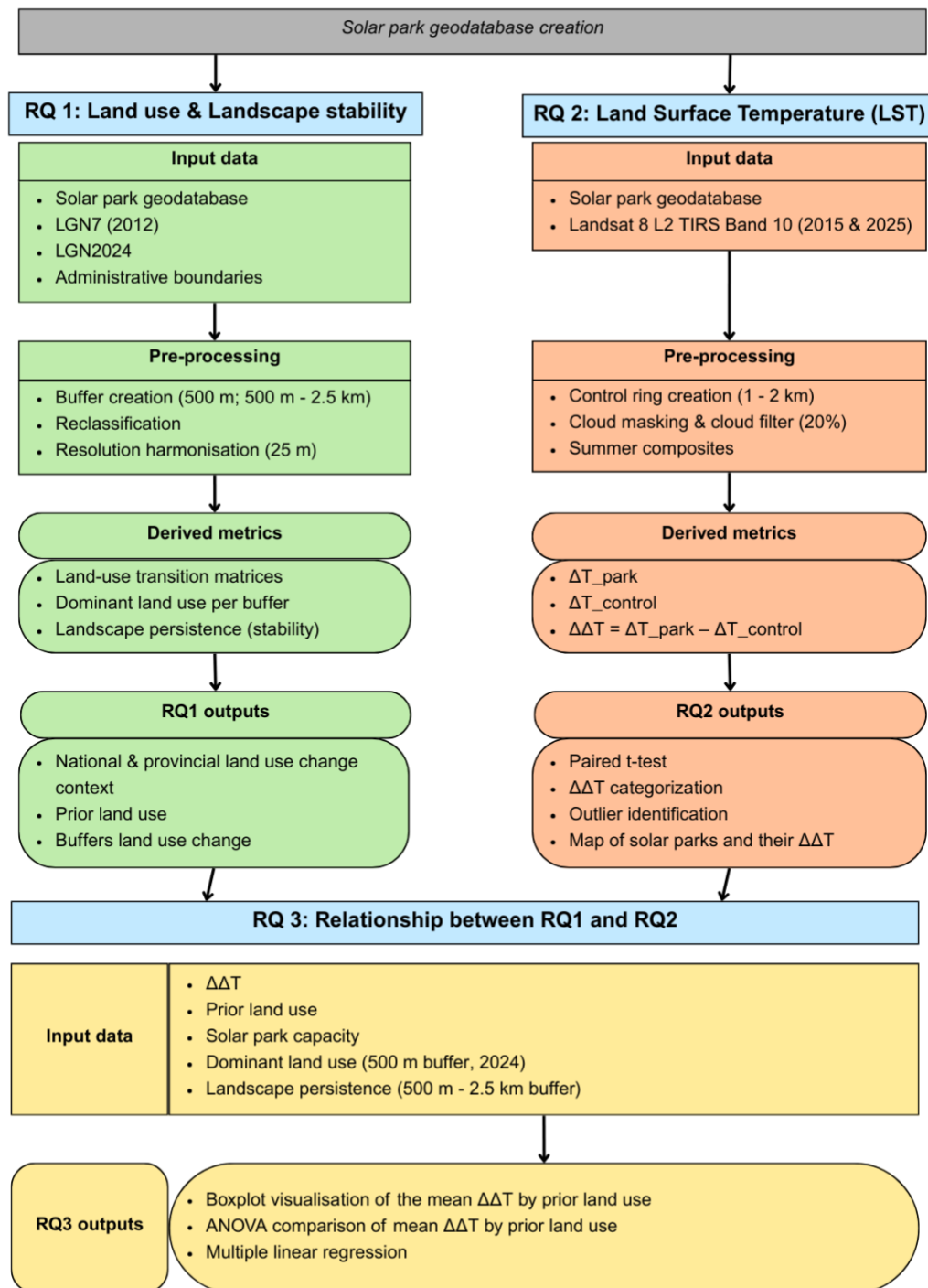


Figure 9: Flowchart of the research process (source: created by the author)

5.1 RQ 1: Land use & landscape stability

Land use change was assessed using The Land Use Database of the Netherlands (Landelijk Grondgebruiksbestand Nederland, LGN) data for the reference year of 2012 (LGN7) and the most recent dataset of 2024 (LGN2024) (Hazeu et al. 2023; 2025). Notably, the reference year precedes the oldest solar park in the study's solar park database (dating from 2016) by four years because LGN jumps from LGN7, representing data from 2012, to LGN2018.

For each solar park, land cover was analysed prior to installation (using the 2012/LGN7 land use data), and land-use change was assessed within two surrounding buffer zones: a 500-meter buffer to capture immediate adjacent changes, and a 2-kilometer buffer extending from 500 meters to evaluate broader landscape stability (see Figure 10 for an example of the buffers). This approach enabled an evaluation of whether solar park siting aligns with policy objectives, such as the solar ladder, and whether the surrounding landscape has remained stable or undergone change. The latter was assessed in reference to national land-use change as captured by the LGN7/LGN2024 datasets, and land-use stability at the provincial level.

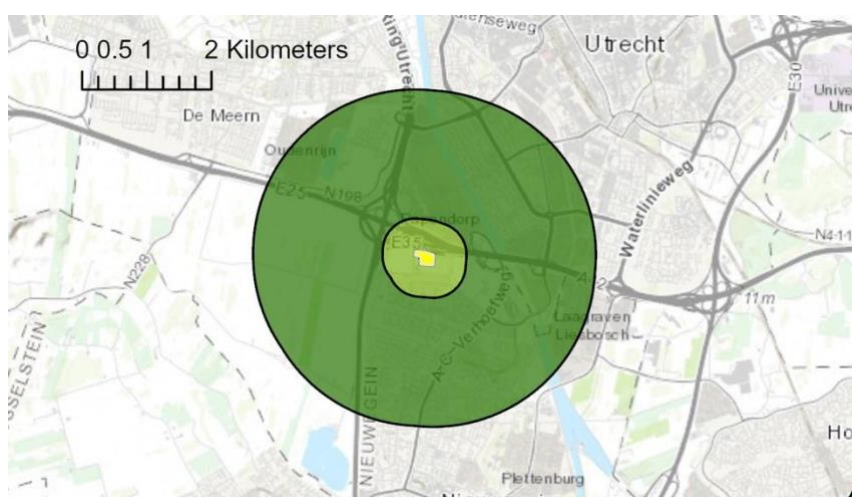


Figure 10: Example of a solar park and its buffer areas of 500 meters (light green) and 2 kilometers (dark green) for the land use analysis (this is solar park 'Galecop' in Utrecht)

Both LGN7 and LGN2024 were generalized into broader land-use categories, following the classification scheme used in the national Landscape Monitor (Monitor Landschap). The Landscape Monitor was developed in 2021 by the Cultural Heritage Agency of the Netherlands on behalf of several ministries (Monitor Landschap, 2023). This classification aggregates more detailed land-use classes—such as individual crop types—into general categories (e.g. agricultural area) that are most characteristic of differences between landscapes (Nieuwenhuizen et al. 2024). The resulting classification consisted of eight aggregated land-use classes. In addition, class 9 (solar parks), which is only present in LGN2024, was retained

as a separate category. This was done to allow explicit identification of land-use transitions toward solar parks and to avoid ambiguity in the reclassification process. An overview of the class conversions is provided in Appendix B, and Table 3 presents the aggregated land-use classes applied in this study.

Table 3: Aggregated land use classes

1 – Agricultural area
2 – Greenhouse horticulture
3 – Orchards
4 – Forest
5 - Water
6 – Urban area
7 – Infrastructure
8 - Nature
9 – Solar parks (LGN2024 only)

The LGN2024 land-use dataset was resampled from 5×5 m cell size to 25×25 m using nearest-neighbour aggregation to match the spatial resolution of LGN7. After reclassification and legend harmonization, the LGN7 and LGN2024 rasters were combined to derive a national-level land-use transition matrix.

For the provincial analysis, administrative boundaries were converted to raster format and aligned with the LGN7 grid. Provincial extents were then combined with the land-use data to generate provincial-level transition matrices.

For the buffer analysis, circular zones of 500 m and 500 m–2.5 km were generated around each solar park. Land-use values from LGN7 and LGN2024 were then extracted for all pixels falling within each buffer. For each park and buffer, the number of pixels transitioning between land-use classes was counted, converted to hectares, and used to construct absolute and relative transition values.

All raster combinations were exported as tabular transition data. Pixel counts (1 pixel = 25×25 m = 625 m²) were converted to hectares and reshaped to construct absolute and percentage transition matrices at the national, provincial, and buffer levels. These transition matrices not only describe changes between land-use classes but also provide the basis for quantifying landscape stability.

Landscape stability is measured using persistence (in %), defined as the proportion of area that remained unchanged between time steps:

$$\text{Persistence} = \frac{\sum \text{area of unchanged land-use classes}}{\text{total area}} \times 100$$

5.2 RQ 2: Land surface temperature (LST)

Land Surface Temperature (LST) is derived from Landsat 8 Collection 2 Level-2 thermal infrared data. The analysis uses the TIRS Band 10 brightness temperature product. Landsat 8 has a 16-day revisit cycle with a morning overpass, and a 100 m thermal resolution. The thermal band was first converted to temperature in Kelvin using the scaling factors supplied with the Level-2 dataset (USGS, 2024).

To ensure consistent cloud-free coverage, all Landsat 8 scenes overlapping the Netherlands during the summer period (1 May – 1 September) were collected for the study years 2015 (prior to the solar parks being built since the oldest year of completion in the solar park database is 2016) and the most recent year of 2025. Clouds and cloud shadows were removed using the pixel quality assessment band (QA_PIXEL), and a pixel-wise median composite was produced. Additionally, a maximum cloud cover filter was set for a maximum cloud cover per used scene of 20%. This yielded a representative summer surface temperature for 2015 and 2025, rather than relying on a single clear-sky day. The resulting true color and thermal images were visually inspected to ensure no significant anomalies and clear skies.

The final composites included 13 scenes from 9 separate dates in 2015 and 18 scenes from 11 separate dates in 2025 that met quality criteria (the scene dates for the composites can be found in Appendix C).

For each park, a control area was defined as a 1 km-wide ring starting 1 km from the park boundary (see Figure 11 for an example). Mean LST values for 2015 and 2025 were calculated for both the solar park area and the control area, allowing temperature changes at the solar park site to be compared with changes in the surrounding landscape.

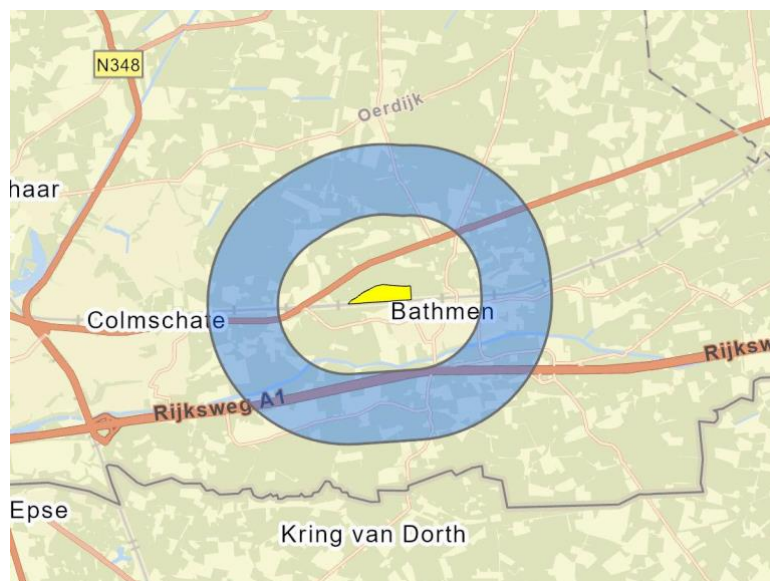


Figure 11: Example of a solar park and its control area for the surface temperature analysis (this is solar park 'Bathmense Landen' in Overijssel)

Mean temperatures in Kelvin were converted to Celsius. For each solar park and its corresponding control ring, the change in mean LST between 2015 and 2025 was calculated. The difference in control area temperature change was then subtracted from the solar park change to calculate the additional warming in solar parks relative to their control areas ($\Delta\Delta T = \Delta T_{\text{park}} - \Delta T_{\text{control}}$).

To test whether the solar parks experienced a statistically significant different change in LST than their surrounding (control) areas, a paired t-test was used to assess whether the temperature changes at the solar parks were consistently larger or smaller than the changes in their corresponding control areas. Because the paired t-test assumes that the paired differences are approximately normally distributed, the distribution of $\Delta\Delta T$ values was visually inspected and found to be roughly symmetric and bell-shaped. The following hypotheses were tested:

$$H_0: \Delta\Delta T = \Delta T_{\text{park}} - \Delta T_{\text{control}} = 0$$

$$(no\ additional\ temperature\ change\ at\ solar\ parks\ relative\ to\ their\ control\ areas) H_1: \Delta\Delta T = \Delta T_{\text{park}} - \Delta T_{\text{control}} \neq 0$$

(solar parks exhibit a systematic difference in temperature change compared to their control areas)

The Pearson correlation coefficient between ΔT_{solar} and $\Delta T_{\text{control}}$ was also calculated to assess the strength of the linear relationship.

5.3 RQ 3: Linking land use change and LST change

Building on the results of the previous research questions, this final analysis integrates land use change (RQ1) and land surface temperature (LST) data (RQ2) to examine how the conversion of land to solar energy production affects local thermal dynamics. Specifically, differences in LST response are assessed in relation to the land-use context in which solar parks were developed, testing whether parks established on different prior land uses (e.g., Agricultural area, Urban area, or Water) exhibit distinct temperature changes.

To evaluate these effects within an integrated modeling framework, the additional warming of each solar park relative to its surrounding control area ($\Delta\Delta T$) was modeled as a function of province prior land use, solar park capacity, park area size, the dominant land use in the immediate 500 m buffer, and the persistence of the broader 500 m–2.5 km landscape

(Figure 12). Interactions between prior land use and the buffer-scale landscape metrics were included to assess how local and broader landscape context modulate thermal responses to solar park development.

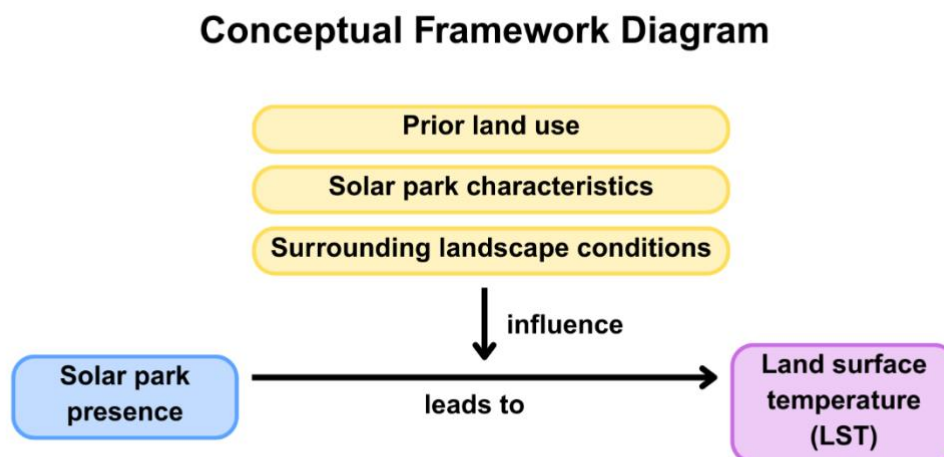


Figure 12: Conceptual framework diagram for the multiple linear regression (source: created by the author)

5.4 Validation and uncertainty analysis

The validation and uncertainty analysis aimed to evaluate the robustness and reliability of the study's results. To assess the robustness of the land surface temperature results, analyses of solar park temperature change ($\Delta\Delta T$) were conducted both including and excluding outliers. Outliers were defined as values exceeding two standard deviations from the mean $\Delta\Delta T$. This approach ensured that extreme observations did not disproportionately influence the statistical results, while still allowing their effects to be evaluated separately.

For the statistical analysis of prior land use in relation to $\Delta\Delta T$, a minimum sample size threshold (minimum N) was applied to avoid unreliable estimates for categories represented by only a few parks. This ensured that comparisons between land-use categories were based on sufficiently robust sample sizes.

For the multiple linear regression model, standard model assumptions were tested, including normality of residuals, homoscedasticity, linearity, and absence of multicollinearity. Diagnostic plots and variance inflation factors (VIF) were examined to assess these assumptions. The regression model was additionally run excluding outliers to evaluate the stability of the estimated coefficients. For statistically significant results, the corresponding sample sizes (N) were reported to provide transparency regarding the robustness of the findings. Finally, the limitations of the data and methodological choices are critically discussed in relation to the interpretation of the results, thereby strengthening the credibility of the study.

6. Results

6.1 RQ1: Land use change results

RQ1: What land-use changes are associated with solar park development, and how do these changes relate to immediate and broader landscape stability and policy goals?

6.2.1. National land use change

Between 2012 and 2024, the national persistence was 86.07%, meaning that the majority of the land area retained the same land-use class. Despite this overall stability, national land-use changes were concentrated in specific classes (Table 4). The full national transition matrix is provided in Appendix D.

In 2024, natural land-use classes (Forests, Nature, and Water) cover approximately 1.5 million hectares, out of the Netherlands' total area of roughly 4.15 million hectares. This represents an increase from 1.25 million hectares in 2012, showing overall growth in these land-use types. Most of this expansion occurred in the Water class, which gained over 90,000 hectares, 75,000 hectares of which came from Agricultural areas. The Forest class gained nearly 12,000 hectares from Agriculture, but also returned about 9,000 hectares back to Agricultural areas elsewhere. The Nature class received almost 70,000 hectares from Agriculture, while only giving back less than half of that to other land-use types.

Agricultural land-use change is dominated by persistence within the agricultural domain and internal reallocation among its classes. The Orchards class shows relatively low persistence, but most of its transitions remain within agricultural land. Similarly, a large portion of land that changes in the Greenhouse horticulture class transitions to Agricultural area. Together, the agricultural land-use classes (Agricultural area, Greenhouse horticulture, and Orchards) cover approximately 2 million hectares, representing a net loss of nearly 200,000 hectares since 2012. Most of this loss—about 120,000 hectares—was converted to the built environment, after accounting for areas transitioning back from built-up land to agriculture.

The built environment (Urban area and Infrastructure) gained nearly 100,000 hectares between 2012 and 2024. Infrastructure exhibits comparatively low persistence, with substantial transitions both into Urban area and Agricultural land. After accounting for areas that transitioned back from built-up land to natural land, the built environment gained roughly 25,000 hectares from natural land-use classes.

Finally, the newly introduced category of solar parks in LGN2024 provides insight into land-use transitions toward this function, although these transitions should be interpreted with caution. The transition matrix includes all solar park developments, whereas the detailed solar park analysis in this study is restricted to installations larger than 4 ha. At the national scale, transitions into solar parks amount to approximately 4,500 ha, representing a minor share of total land-use change compared to other transition pathways. Of the land converted to solar parks, approximately 3,000 ha originated from Agricultural area, 700 ha from Urban area and 200 ha from Water.

Table 4: *Most significant national land-use transitions (in %) (2012-2024)*

Source land use (2012)	Dominant transition patterns (2024)
Agricultural area	~87% unchanged; ~4% to Water; ~3% to Nature ~3% to Urban area; ~2% to Infrastructure
Greenhouse horticulture	~84% unchanged; ~9% to Agricultural area; ~3% to Urban area
Orchards	~66% unchanged; ~25% to Agricultural area; ~3% to Water, ~ 2% to Nature and ~2% to Infrastructure
Forests	~79% unchanged; ~7% to Nature; ~5% to Infrastructure; ~4% to Urban area; ~3% to Agricultural area
Water	~97% unchanged; ~1% to Nature; ~1% to Infrastructure
Urban area	~83% unchanged; ~10% to Infrastructure; ~3% to Agricultural area; ~2% to Water; ~1% to Nature; ~1% to Forests
Infrastructure	~51% unchanged; ~20% to Urban area; ~20% to Agricultural area; ~6% to Water; ~3% to Forests; ~1% to Nature
Nature	~71% unchanged; ~12% to Agricultural area; ~8% to Forests; ~7% to Water; ~2% to Infrastructure; ~1% to Urban area

6.2.2. Provincial land use change

Persistence varied across provinces, ranging from 80.6% in Utrecht to 87.1% in Noord-Holland (Table 5).

Table 5: *Provincial persistence metrics. Persistence is expressed as the percentage of land retaining the same land-use class between 2012 and 2024.*

Province	Persistence (%)
Noord-Holland	87.10
Drenthe	85.54
Fryslân	85.41
Overijssel	84.98
Groningen	84.22
Zeeland	83.97
Flevoland	83.43
Noord-Brabant	82.98
Zuid-Holland	82.21

Gelderland	81.59
Limburg	81.04
Utrecht	80.57
<i>National</i>	<i>86.07</i>

6.2.3. Prior land use

Table 6 shows the dominant prior land-use classes underlying the solar parks included in the database. For solar parks that overlap multiple land-use classes in the 2012 land-use map, each park was assigned to the land-use class covering the largest share of its area. The table therefore reflects the primary land-use type from which each solar park was developed.

Table 6: *Number of parks per primary land use
(where most of the solar park's area is located in this land use type)*

Dominant land use class	Number of parks
Agricultural area	153
Urban area	34
Water	9
Orchards	4
Nature	4
Forest	1
Infrastructure	1
Total	206

The table shows that most solar parks in the study database (all >4 ha) were built on primarily agricultural area (153 parks, 74.27%), with 34 on urban area (16.5%) and 10 on natural areas (Water, Nature, Forest) (6.8%). The surprisingly large number of solar parks located on previously urban land are predominantly situated on former landfill sites.

6.2.4. Persistence and dominant land use around the solar parks

An analysis of dominant land-use transitions suggests that, in most cases, the prevailing land-use class in areas surrounding solar parks remained unchanged over time. Dominant land use is defined as the land-use class occupying the largest proportion of each buffer area. Within the 500 m buffers, Agricultural area was the dominant class for 167 parks, followed by Urban area (26), Nature (5), Forest (4), Water (3), and Orchards (1).

Of the 206 parks analysed, 190 retained the same dominant land-use class in both the 500 m and 2 km buffers between 2012 and 2024. Only 16 parks experienced a change in dominant land use, primarily involving transitions from Agricultural area to Urban area within

the 500 m buffer (5 cases). Detailed land-use transitions for these 16 parks are presented in Appendix G.

The average persistence immediately adjacent to solar parks (500 m buffer) is 79.45%, lower than the national average of 86.07%, reflecting the influence of solar park development on the surrounding landscape (note that the parks themselves are excluded from this buffer).

In the broader 500 m–2 km buffer, persistence rises to 83.84%, closer to the national average. However, the difference is more notable in this broader landscape because, unlike immediately adjacent to the park, some level of change would be less expected. This suggests that solar parks are often established in relatively dynamic areas.

Persistence in the immediate 500 m buffers around solar parks varies across provinces, ranging from 72.7% in Limburg to 81.58% in Gelderland, with Noord-Holland showing the largest gap between the 500 m buffer (77.01%) and the provincial average (87.10%) (Table 7). This suggests that in some provinces, solar parks are situated in areas with more dynamic land-use patterns. In the broader 2 km buffers, persistence is generally closer to or even above provincial averages.

Table 7: Average persistence in buffers around solar parks by province (2012-2024)

Province	Persistence (%)	Persistence 500m	Persistence 2 km	N
Noord-Holland	87.10	77.01%	84.25%	13
Drenthe	85.54	80.17%	83.77%	24
Fryslân	85.41	79.55%	84.53%	17
Overijssel	84.98	81.42%	86.09%	20
Groningen	84.22	79.50%	83.34%	21
Zeeland	83.97	81.11%	84.79%	11
Flevoland	83.43	75.27%	83.58%	11
Noord-Brabant	82.98	80.32%	82.89%	20
Zuid-Holland	82.21	77.41%	83.57%	11
Gelderland	81.59	81.58%	83.43%	39
Limburg	81.04	72.66%	82.02%	7
Utrecht	80.57	76.95%	83.29%	12
<i>National</i>	<i>86.07</i>			<i>206</i>

6.2.5. Land-use transitions around the solar parks

In addition to assessing the prevailing land use in the buffers, all land-use transitions within the 500 m and 2 km buffers were analyzed to assess how land use change around solar park development compares to national trends. In the 500 m buffers, which collectively cover approximately 33,574 hectares, the largest absolute changes occurred from Agricultural area to Urban area (~907 ha), followed by transitions from Agricultural area to Water (~783 ha),

Nature (553 ha), and Infrastructure (~544 ha). All of these transitions exceed national averages (Table 4). Table 8 shows the most significant transitions for land uses covering more than 1,500 ha, as smaller classes would yield misleading percentages and are negligible in absolute terms. While transitions toward Agricultural area, originating from Water or Urban area, are also evident and more than the national average, their absolute magnitude is smaller and less influential, given that Agricultural area already occupies approximately one-third of the total 500 m buffer area. Urban area occupies ~5500 ha and Water ~1800 ha.

Table 8: Most significant land-use transitions in the 500 m buffers (aggregated) (in %) (2012-2024)

Source land use (2012)	Dominant transition patterns (2024)
Agricultural area	~84% unchanged; ~4% to Urban area; ~4% to Water ~3% to Nature; ~3% to Infrastructure;
Forests	~72% unchanged; ~9% to Nature; ~6% to Agricultural area; ~5% to Urban area; ~5% to Infrastructure
Water	~89% unchanged; ~4% to Infrastructure; ~2% to Agricultural area; ~2% to Nature
Urban area	~76% unchanged; ~8% to Agricultural area; ~7% to Infrastructure; ~3% to Water; ~3% to Nature

Broader 2 km buffers, totaling roughly 457,083 ha, show transition patterns more similar to the national transition patterns, especially for Agricultural area, which accounts for more than half of the 2 km buffers' area. Nevertheless, there is a higher relative transition from the other land use classes to Agricultural than in the national land use transitions (from Forest, Urban area, Infrastructure, and Nature) (Table 9). Finally, similar to the 500 m buffers, there are distinctly higher transition rates to Urban area from the natural land use classes (Forest, Water, and Nature).

Table 9: Most significant land-use transitions in the 2 km buffers (aggregated) (in %) (2012-2024)

Source land use (2012)	Dominant transition patterns (2024)
Agricultural area	~87% unchanged; ~3% to Urban area; ~3% to Water ~3% to Nature; ~2% to Infrastructure;
Forest	~74% unchanged; ~7% to Nature; ~6% to Urban area; ~5% to Agricultural area; ~5% to Infrastructure
Water	~94% unchanged; ~5% to Urban area; ~3% to Infrastructure
Urban area	~83% unchanged; ~9% to Infrastructure; ~4% to Agricultural area; ~2% to Water
Infrastructure	~50% unchanged; 22% to Agricultural area; ~19% to Urban area; ~6% to Water; ~2% to Forest

Nature	~65% unchanged; ~16% to Agricultural area; ~9% to Forest; ~7% to Water; ~2% to Urban area; ~2% to Infrastructure
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6.2 RQ2: Land surface temperature results

RQ2: How do solar parks influence local land surface temperatures?

The mean temperature change between 2015 and 2025 was $+0.469^{\circ}\text{C}$ in solar park areas, compared to -0.216°C in control areas. This corresponds to an additional mean warming ($\Delta\Delta T$) of 0.685°C in solar park areas relative to their controls. The difference in temperature change between the solar park and control areas is statistically significant (paired t-test, $p < 0.01$).

The Pearson correlation coefficient between ΔT_{solar} and $\Delta T_{\text{control}}$ was 0.772, indicating a strong positive linear relationship between temperature changes in the solar parks and their control areas, as is expected.

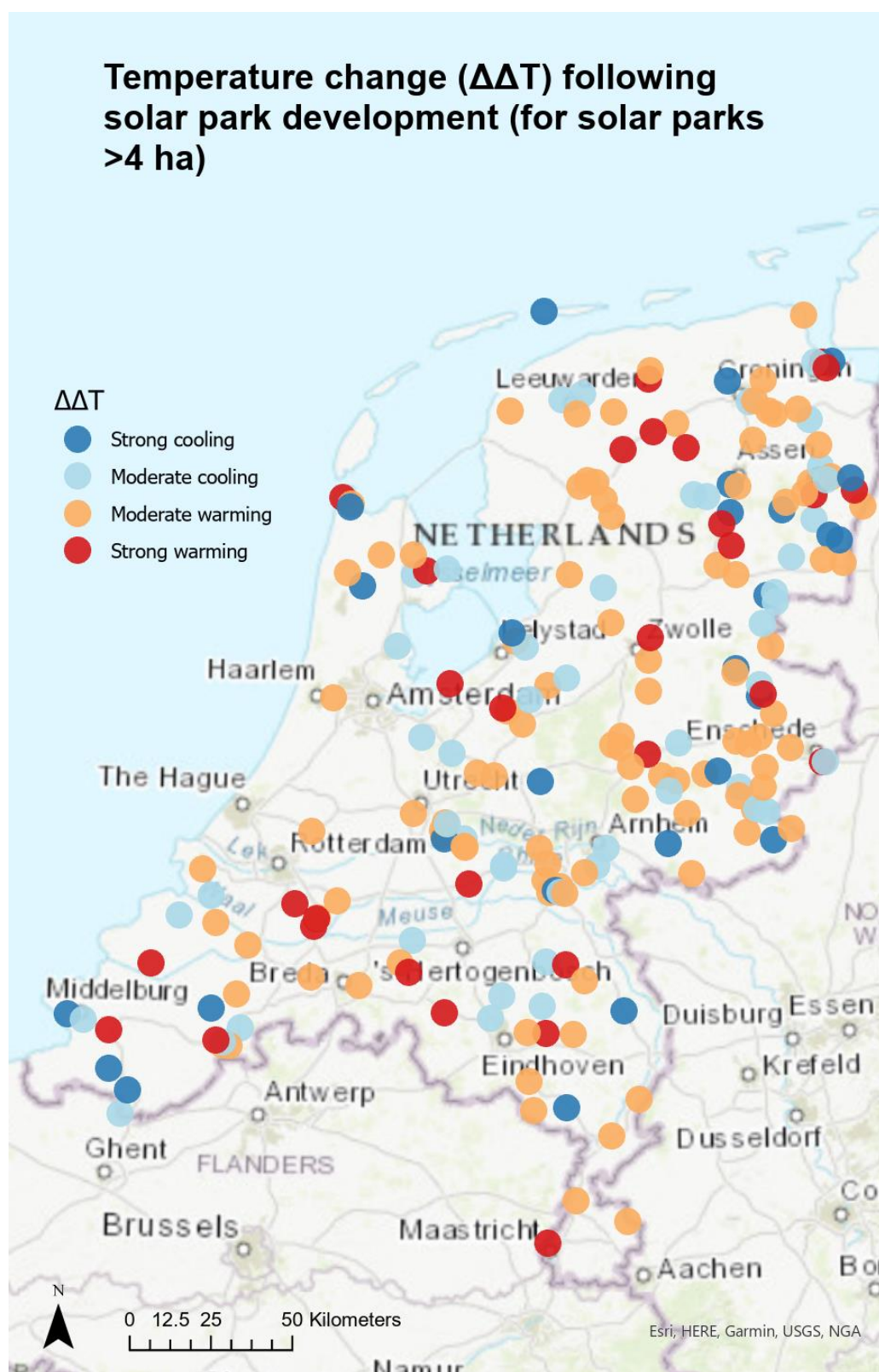
The standard deviation of $\Delta\Delta T$ is 2.070°C . This suggests a large variation in the additional temperature change on solar park sites. The minimum $\Delta\Delta T$ is -7.881°C , and the maximum $\Delta\Delta T$ is $+7.147^{\circ}\text{C}$. If outliers are defined as > 2 standard deviations, there are six cold outliers and four warm outliers.

Excluding outliers ($N=196$) does not alter the results. Solar park areas still exhibit a significantly greater temperature increase than control areas ($p < 0.001$), with an average additional warming of 0.74°C . The temperature changes in solar parks and control areas remain strongly correlated ($r = 0.792$).

The solar parks can be categorized into one of four categories based on their $\Delta\Delta T$ value (Table 10). Values lower than or equal to one standard deviation from the mean (≤ -1.522) are categorized into strong cooling, values in between one standard deviation below the mean and 0 ($-1.522 < \Delta\Delta T < 0$) are categorized into moderate cooling, values higher than 0 and lower than one standard deviation above the mean ($0 \leq \Delta\Delta T < +2.892$) are categorized into moderate warming, and values equal to or higher than one standard deviation above the mean ($\geq +2.892$) are categorized as strong warming. Figure 13 shows the spatial distribution of the temperature change categories.

Table 10: *Descriptive statistics by category*

Category	Count	Percentage of total (N=206)
Strong cooling	28	13.59
Moderate cooling	48	23.30
Moderate warming	100	48.54
Strong warming	30	14.56



6.3 RQ3: Relationship between land use change and LST change

RQ3: To what extent are variations in land surface temperature within Dutch solar parks associated with prior land use and surrounding landscape stability?

Descriptive statistics for $\Delta\Delta T$ by prior land-use class are presented in Table 11 and visualized in Figure 14. Among land-use classes with at least five solar parks, $\Delta\Delta T$ varied, with solar parks established on water areas showing the highest mean $\Delta\Delta T$ (mean = 3.4°C). In contrast, parks on agricultural grass, arable or horticultural crops, corn, and urban areas displayed lower and broadly similar mean $\Delta\Delta T$ values (means ranging from -0.3°C to 0.82°C).

Table 11: Descriptive statistics by prior land use for land use classes with > 5 solar parks

Dominant prior LU class	N	Mean $\Delta\Delta T$	Standard deviation
Agricultural area	153	0.654°C	2.10°C
Urban area	34	-0.028°C	2.42°C
Water	9	3.422°C	1.28°C
Total	204		

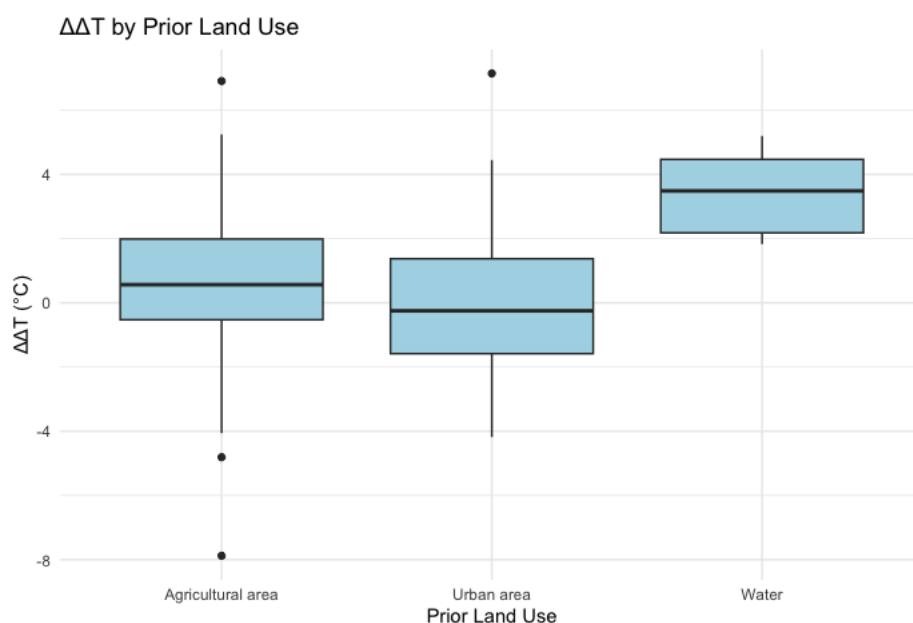


Figure 14: Distribution of $\Delta\Delta T$ by prior land-use class (classes with ≥ 5 solar parks)

An ANOVA indicates that the additional warming effect ($\Delta\Delta T$) differs significantly between dominant prior land-use classes ($F = 9.36$, $p < 0.001$). Post-hoc Tukey tests show that solar

parks located in areas previously classified as Water exhibit significantly higher additional warming compared to both Agricultural areas ($p < 0.001$) and Urban areas ($p < 0.001$). No statistically significant difference is observed between Agricultural and Urban prior land-use classes ($p = 0.21$). Excluding outliers does not alter the conclusions; the effect of prior land use remains statistically significant ($p < 0.001$), and the relative differences between land-use classes are unchanged.

A multiple linear regression was conducted to examine potential drivers of additional warming ($\Delta\Delta T$) at solar parks, including prior land use, dominant land use in the 500 m buffer, landscape persistence in the 500 m–2 km buffer, province, solar park capacity, and solar park area. With the outlier $\Delta\Delta T$ s included ($N=206$), the residuals showed deviations from normality, but after removal of these outliers ($N=196$), the assumption of normality of residuals was met. The model explained a modest proportion of the variance in $\Delta\Delta T$ ($R^2 = 0.262$, Adjusted $R^2 = 0.154$, $F = 2.42$, $p < 0.001$) but revealed significant relationships between prior land use, dominant land use in the 500 m buffer, and province on local temperature changes. Area size had small effects, while solar park capacity and persistence in the surrounding landscape (500m – 2km) were not significant predictors.

Solar park areas where the prior land use was Water are associated with substantially higher $\Delta\Delta T$ (estimate = 2.95, $p < 0.001$, $N = 9$). Parks located in urban-dominated 500 m buffers experience significantly lower $\Delta\Delta T$ compared to other land-use types (estimate = -1.13, $p = 0.007$, $N = 26$). Among provinces, Limburg (estimate = 2.01, $p = 0.016$, $N = 7$) and Noord-Holland (estimate = 1.61, $p = 0.011$, $N = 13$) are associated with higher $\Delta\Delta T$, independent of park characteristics. Solar park area also shows a small but significant positive effect on $\Delta\Delta T$ (estimate = 3.66×10^{-6} , $p = 0.041$). Other predictors—including prior land use, the broader 500 m–2 km persistence, and most provinces—are not statistically significant, indicating that the warming effect is primarily shaped by immediate land-use context, regional factors, and the physical size of the park.

7. Discussion

7.1 Land-use context and solar park development

National land-use trends between 2012 and 2024 show a gradual shift from agricultural land toward more natural land uses, while the built environment (particularly Infrastructure) is the most dynamic category. Against this relatively stable national background (86.07% persistence), solar parks account for approximately 4,500 ha—just over 0.1% of the Dutch territory. Although modest in national terms, their siting pattern is revealing: 153 parks were developed on former Agricultural area (~3,000 ha), followed by 34 on Urban area (~700 ha) and 9 on Water (~200 ha). This confirms that solar park development has primarily taken place on farmland, consistent with findings by Becker et al. (2025).

From a policy perspective, this contradicts the solar ladder policy, which prioritizes rooftops and built-up areas before rural land, explicitly aiming to protect agricultural and natural landscapes. However, most solar parks in this dataset were developed before the policy was introduced in 2020 and reinforced in 2023. The observed siting pattern therefore primarily reflects a pre-solar ladder development phase characterized by stronger market dynamics and limited spatial steering. To properly assess policy impact, future research could isolate parks developed after 2020. Moreover, the solar ladder applies to solar PV deployment as a whole, whereas this study focuses only on ground-mounted parks larger than 4 ha. A comprehensive evaluation would require integrating rooftop installations and smaller projects.

Landscape stability analysis provides additional context. Persistence in the 500 m buffers (79.45%) is substantially lower than both national and provincial averages, indicating that solar parks are generally located in relatively dynamic landscapes characterized by ongoing land-use transitions and multifunctionality. At the 2 km scale, persistence rises to 83.84%, approaching national levels, yet remains slightly below the national average. This suggests that solar parks are overrepresented in areas already undergoing transformation rather than in highly stable agricultural or natural cores. Provincial variation confirms that this context differs regionally. However, provincial comparisons must be interpreted cautiously where sample sizes are small. For example, the pronounced gap between provincial and buffer persistence in Limburg is based on only seven parks, limiting statistical robustness. Larger regional samples would yield more reliable estimates of contextual stability.

Transition patterns reinforce this interpretation. Within the 500 m buffers, agricultural land is the main donor class, with above-average transitions toward Urban area and Infrastructure. Urbanisation signals are particularly strong in the immediate surroundings

(500-m buffers) compared to national trends. The broader 2 km buffers more closely resemble national transition dynamics, but still show elevated transitions from natural land uses toward Urban area. This reinforces that solar parks are frequently located in peri-urban or transitional landscapes where multiple land-use pressures converge. A further study into the potential causality of the spatial association between solar (park) development and urbanization is beyond the scope of this research.

Overall, the findings suggest that solar parks are predominantly introduced into landscapes already experiencing change, rather than introducing disruption into highly stable systems. From a landscape stability perspective, the intervention may therefore be less disruptive than if parks were systematically located in highly persistent agricultural or natural cores. At the same time, the continued reliance on agricultural land as the most common siting location highlights an ongoing tension between renewable energy deployment and farmland protection—one of the central concerns underlying the solar ladder policy. And if solar park development goes hand in hand with urbanization, the environmental implications of the solar park development could be amplified.

This landscape context is relevant for interpreting land surface temperature (LST) change examined hereafter. Land-use composition and stability shape vegetation cover, soil moisture, and surface characteristics, which in turn influence local climate regulation. However, a key limitation is that this study does not account for management practices within solar parks. As shown by Schotman et al. (2021), vegetation structure and biodiversity outcomes vary substantially depending on site management. Agri-PV systems, for example, differ fundamentally from single-use ground-mounted parks in their land-use integration and ecological effects.

7.2 Thermal impact of solar parks

This study found a statistically significant mean additional warming effect of $+0.685^{\circ}\text{C}$ in Dutch solar parks relative to paired control areas over the 2015–2025 period. These results contrast with large-scale studies reporting dominant daytime cooling effects of photovoltaic (PV) installations (Xu et al. 2024; Zhang & Xu, 2020).

For example, Xu et al. and Zhang & Xu found that large solar farms often exhibit daytime cooling relative to adjacent land, although nighttime warming may occur. Importantly, both studies do report that cooling effects decrease with latitude and are generally weaker in temperate climates. The Netherlands ($\sim 52^{\circ}\text{N}$) lies at a higher latitude than the sites included in these global assessments. Lower incoming solar radiation and frequent cloud cover may reduce the relative contribution of PV conversion efficiency to local cooling, while allowing increased sensible heat flux due to high absorption of solar radiation to exert a stronger

influence (Heusinger et al., 2020). Panel tilt, spacing, and park management practices are factors not available for this study, but they may further influence thermal outcomes through shading, vegetation cover, and local airflow. The dominance of agricultural prior land use may also limit cooling, as these areas already exhibit moderate albedo and evapotranspiration. At the same time, there is strong variability across sites in this study, which underscores that there is no uniform solar park temperature effect. ΔT values range from pronounced cooling (-7.9°C) to substantial warming ($+7.1^{\circ}\text{C}$), and approximately 37% of parks exhibit net cooling relative to controls, while 63% show net warming. This heterogeneity reinforces findings from previous research that PV-induced temperature responses are highly context-dependent (Xu et al., 2024; Zhang & Xu, 2020).

Methodologically, the strong correlation between solar parks and their paired control areas supports the robustness of the comparative design. Both treatment and control sites are embedded within the same regional climate trends, indicating that the ΔT metric effectively isolates relative differences rather than reflecting broader climatic warming. Moreover, the persistence of the warming signal after removing statistical outliers suggests that the observed effect is not driven by a small number of extreme cases.

Uncertainties in satellite-derived land surface temperature (LST) measurements should nevertheless be acknowledged. Previous studies have debated whether standard emissivity assumptions bias LST retrieval over photovoltaic arrays. Xu et al. (2024) argue that emissivity overestimation may lead to underestimation of true surface temperatures over solar panels, whereas Zhang & Xu (2020) suggest that panel geometry and cavity effects may compensate for such differences. Future research could incorporate an emissivity correction for solar park area.

Overall, rather than contradicting previous research in other areas of the world, these results extend it by providing evidence from a climatic and spatial context that has been underrepresented in earlier studies. The pronounced variability across sites further emphasizes that thermal impacts of solar parks are not uniform and should be interpreted within their broader environmental and spatial setting.

Future research could further explore land surface temperature dynamics in the Netherlands using MODIS LST products, which offer higher temporal resolution (up to four daily overpasses and 8-day composites) and would allow for explicit analysis of daytime and nighttime differences. However, the coarser spatial resolution of MODIS (1 km) compared to Landsat (100 m) introduces potential mixed-pixel issues, particularly in the Netherlands, where only a limited number of solar parks exceed 400 ha—the minimum size threshold applied by Xu et al. (2024) to reduce mixed-pixel contamination. Given that most Dutch solar parks are substantially smaller, MODIS-based analyses may obscure site-level variability and

dilute park-specific thermal signals. A combined multi-sensor approach could provide complementary insights while balancing spatial and temporal resolution trade-offs.

7.3 Influence of prior land use and local landscape

The analysis of the relationship between additional warming ($\Delta\Delta T$) at solar parks and land-use dynamics shows that local thermal responses are influenced by prior land use and the immediate landscape context. Solar parks established on former water bodies exhibit the highest mean $\Delta\Delta T$ ($\sim 3.4^\circ\text{C}$), significantly exceeding that of parks on agricultural or urban land, which show modest warming or slight cooling. While the sample size for water ($N=9$) and other minor land-use categories is limited, this pattern aligns with expectations: water is typically cooler than surrounding land, so replacing it with PV arrays naturally produces greater relative warming. Nevertheless, a larger sample size for each prior land-use category would be desirable to further investigate these relationships, as only three classes (Agricultural area, Urban area, and Water) had $N > 5$ and could be included in this study's ANOVA. Interestingly, Xu et al. (2024) also found LST impacts to be very dependent on prior land use, with barren or grassland sites experiencing the most cooling, while cropland sites showed weaker cooling or nighttime warming. This study found no significant difference $\Delta\Delta T$ between Agricultural area and other prior land uses (except Water), but future research could refine land-use classifications (e.g., cropland, grassland, barren) to assess these effects more robustly and test whether patterns observed globally hold in the Dutch context.

Immediate surroundings also influence $\Delta\Delta T$. Parks located within urban-dominated 500 m buffers experience significantly lower additional warming ($\sim 1.1^\circ\text{C}$), likely because built-up areas are already thermally elevated, reducing the relative impact of PV installation. This observation has practical and policy relevance: developing solar parks near urban areas not only aligns with the solar ladder's preference for built-up land but may also minimize additional thermal stress. However, the study found that solar parks frequently coincide with higher transition rates to built-up area, raising the possibility that solar development could overlap with ongoing urbanization, which would in turn cause more heat stress again. However, this is a relationship observed here but not formally established as causal.

Broader landscape persistence (500 m–2 km) did not significantly affect $\Delta\Delta T$, suggesting that local thermal outcomes are primarily shaped by immediate land-use context rather than larger-scale landscape stability. Some provincial effects were apparent: Limburg and Noord-Holland exhibited elevated $\Delta\Delta T$ independent of park characteristics. Notably, both provinces show the largest gaps between provincial persistence and 500-m buffer persistence (~ 10 percentage points), indicating that locally dynamic landscapes may amplify

thermal responses. Park area has a very small positive effect on $\Delta\Delta T$, while capacity does not. This positive effect of area size is consistent with literature, but its magnitude is less pronounced in this study than others (e.g. Xu et al., 2024).

Several limitations should be acknowledged when interpreting the results of this study. First, the use of land-use transition data from 2012–2024 alongside temperature changes from 2015–2025 is not perfectly aligned temporally, which may limit the precision of linking observed warming to specific landscape changes.

Second, the observational design of this study inherently limits causal inference: temperature changes attributed to solar parks result from multiple interacting factors, including factors not taken into account in this study such as climate variability, local management practices, panel design, and vegetation cover. Additionally, the geometries of the solar park polygons themselves are imperfect, occasionally misrepresenting park footprints and thus area-based calculations. Finally, the aggregated nature of the study highlights the high site-specific variability: while some parks exhibit net cooling, others show substantial warming, emphasizing that generalized conclusions must be treated cautiously and that localized assessments are essential.

The ecological and societal implications of this research area are considerable. Localized warming in and around solar parks may influence microclimate conditions, affecting soil processes, vegetation growth, and habitat quality. These effects can interact with ongoing climate change, biodiversity loss, and increasing land-use pressures associated with the energy transition, potentially amplifying stress in sensitive areas. Parks on former water bodies or in highly dynamic landscapes are particularly prone to elevated warming, whereas parks in urbanized or structurally stable settings contribute less additional thermal stress. From a policy perspective, these results emphasize the importance of considering prior land use and immediate landscape context in solar park siting. Promoting multifunctional land-use strategies, such as agri-PV systems or development on previously urbanized or less ecologically sensitive land, can help balance renewable energy expansion with climate-sensitive land management.

Overall, these findings extend previous global studies (Xu et al., 2024; Zhang & Xu, 2020) by providing evidence from a high-latitude, temperate context previously underrepresented in LST assessments. The pronounced variability across sites underscores that PV-induced thermal responses are highly context-dependent, shaped by prior land use, immediate landscape configuration, and other factors.

8. Conclusion

The expansion of solar parks in the Netherlands as part of the energy transition raises questions about spatial planning in a densely used landscape. Through land use and land cover change, solar parks interact with broader landscape development and alter local surface energy balances. The local impacts of solar parks on land surface temperatures (LST) remain understudied in the Dutch context. This study examined land-use change and landscape stability around Dutch solar parks and quantified associated variations in local LST to better understand how solar parks shape their surrounding environment. A geodatabase of 206 Dutch solar parks ≥ 4 ha was derived from existing datasets and enriched with key characteristics to support analysis of land-use and LST changes. Land-use change associated with solar parks was assessed using national land-use datasets prior to and post-solar park development, examining changes within and around each park to quantify local and broader landscape stability. Land surface temperature (LST) effects were evaluated using Landsat 8 thermal imagery, comparing temperature changes at the parks with surrounding control areas to detect temperature change attributable to the installations. These temperature effects were related to prior land use of the park sites. Finally, the relationship between solar park characteristics, land use change and temperature change was explored through a multiple linear regression (MLR). Results show that most solar parks were developed on agricultural land, contradicting national policy. Average landscape stability adjacent to (500 m around) the parks and in the broader landscape (500 m – 2.5 km) around the parks is consistently lower than the national landscape stability, but there is large provincial variety. The LST analysis shows a mean additional warming in solar park areas of 0.69°C , but there is a large variation in this additional temperature change, with 77 parks showing relative cooling. Results of the MLR indicate that solar parks on water consistently exhibit higher additional warming, while parks located within urban-dominated 500 m buffers show slight cooling. Overall, the model explains only a modest portion of $\Delta\Delta T$ variability, highlighting that local thermal responses are highly context dependent. These findings suggest that solar parks primarily integrate into already dynamic landscapes rather than highly stable areas, and that prior land use and immediate landscape context play a key role in local thermal impacts. Implications include potential effects on microclimate, soil processes, vegetation growth, and habitat quality, particularly in sensitive or highly dynamic areas. Solar parks should be preferentially located in already dynamic but ecologically low-value areas, while avoiding thermally sensitive surfaces such as open water and vulnerable soils. Environmental assessments should include microclimate evaluation, particularly for larger parks, to anticipate cumulative warming effects. Aligning national policy goals with

provincial implementation and embedding thermal and landscape stability criteria into permitting processes, would help ensure that solar expansion contributes to the energy transition without undermining local environmental resilience. Future research could examine technical design specifics, park management practices, and day/night temperature variation, and expand the analysis to include smaller parks and rooftop PV to guide climate-sensitive and ecologically informed solar park planning in the Netherlands.

9. Use of Generative AI Statement

During the writing of this thesis, the author used ChatGPT (OpenAI) in order to receive feedback on grammar, writing structure, and coding-related issues. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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IEA (2024-b), World Energy Balances (Figure 2)

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11. Appendices

Appendix A: Solar park sources consulted

Solar Park Name	Website URL
Agri-PV Royal Pride Middenmeer	groenleven.nl/projecten/agri-pv-royal-pride-middenmeer/
Belvedere	bodemzorglimburg.nl/zonneparken/belvedere
BurgerZonnepark Vloelvelden	ecloz.nl/burgerzonnepark/
Cosun Solar Park Puttershoek	greenchoiceintegrated.nl/projecten/cosun-solar-park/
De Groene Energie Corridor	dgec.nl
Drijvend zonnepark Beilen	groenleven.nl/projecten/beilen-drijvend-zonnepark/
Drijvend Zonnepark Bomhofsplas	groenleven.nl/projecten/drijvend-zonnepark-bomhofsplas/
Drijvend Zonnepark de Uivermeertjes	groenleven.nl/projecten/drijvend-zonnepark-de-uivermeertjes/
Drijvend zonnepark Kloosterhaar	groenleven.nl/projecten/drijvend-zonnepark-kloosterhaar
Drijvend Zonnepark Lippe Gabriëlsplas	groenleven.nl/projecten/drijvend-zonnepark-libbe-gabriëlsplas/
Drijvend zonnepark Nij Beets	groenleven.nl/projecten/drijvend-zonnepark-nij-beets/
Drijvend Zonnepark Sekdoornse Plas	groenleven.nl/projecten/drijvend-zonnepark-sekdoornse-plas/
Drijvend zonnepark Sellingen	groenleven.nl/projecten/drijvend-zonnepark-sellingen-met-batterijopslag
Drijvend zonnepark Tynaarlo	groenleven.nl/projecten/drijvend-zonnepark-tynaarlo/
Energiepark Haringvliet Zuid	haringvlietzuid.vattenfall.nl/energiepark-haringvliet/
Energietuin Assen-Zuid	windunie.nl/projecten/energietuin-assen-zuid
Sinnefjild de Ekers	chintsolar.com/nl/projecten/sinnefjild-de-ekers/
Solar Campus Purmerend	solarcampuspurmerend.nl
Solar Carport Biddinghuizen	novar.nl/projecten/solar-carport-biddinghuizen/
Solarpark De Kwekerij	nlsolarparkdekwekerij.nl
Solarvation Lelystad	zonhub.com/project/solarvationlelystad2/
Sunport Delfzijl	zonnepanelenplanet.nl/zonneparken/groningen/zonnepark-delfzijl/
Vlowijker Zonneweide	chintsolar.com/nl/projecten/vlowijker-zonneweide/
Wervershoof	solarmagazine.nl/nieuws-zonne-energie/i24268/hoogheemraadschap-hollands-noorderkwartier-neemt-zonnepark-wervershoof-in-gebruik
Zonneakker Voorst	solarmagazine.nl/nieuws-zonne-energie/i22368/familie-gooiker-neemt-grootste-particuliere-zonnepark-van-nederland-in-gebruik
Zonnecarport TT Circuit Assen	groenleven.nl/projecten/zonnecarport-tt-circuit-assen/
Zonnepark 't Goy	houtensnieuws.nl/lokaal/natuur-en-milieu/362832/zonnepark-t-goy-levert-aan-4500-huishoudens-groene-energie-in-r
Zonnepark 't Rikkerink	twence.nl/projecten/zonnepark-t-rikkerink
Zonnepark 't Zonneveld	lcenergy.nl/projecten/zonnepark-t-zonneveld/
Zonnepark Aadijk	solarmagazine.nl/nieuws-zonne-energie/i8353/solarenergyworks-gestart-met-bouw-zonnepark-aadijk-van-120-808-zonnepanelen

Zonnepark Aadijk II	profinrg.nl/agri-pv-in-almelo/
Zonnepark Aarle-Rixtel	sun-projects.nl/nl/project/laarbeek-aarle-rixtel
Zonnepark Abdissenbosch	landgraaf.nl/zonnepark-abdissenbosch
Zonnepark Afleidingskanaal	ixzon.nl/lochem/
Zonnepark Ameland	eneco.nl/over-ons/wat-we-doen/duurzame-bronnen/zon/zonnepark-ameland/
Zonnepark Amstelwijck	hvcgroep.nl/over-ons/onze-groene-projecten/zonnepark-amstelwijck-dordrecht
Zonnepark Andijk	chintsolar.com/nl/projecten/zonnepark-andijk/
Zonnepark Apeldoorn	https://profingreen.com/nl/project/zonnepark-apeldoorn/
Zonnepark Apeldoorn Bloemenkamp	duurzaam-beleggen.nl/2021/09/10/investeer-in-zonnepark-apeldoorn-bloemenkamp/
Zonnepark Appelscha	groenleven.nl/projecten/zonneparken-ooststellingwerf/
Zonnepark Armhoede	tpsolar.nl/armhoede/
Zonnepark Arrisveld	werkenaanwinterswijk.nl/duurzaamheid/nieuws/2181016.aspx
Zonnepark Attero Wijster	groenleven.nl/projecten/zonnepark-attero-wijster/
Zonnepark Avri	novar.nl/projecten/zonnepark-avri-geldermalsen/
Zonnepark Azewijn	wienerberger.nl/over-wienerberger/duurzaamheid/solarpark-azewijn.html
Zonnepark Batendijk	ixzon.nl/batendijk/
Zonnepark Bathmense Landen	zonhub.com/projecten/bathmenselanden/
Zonnepark Bathpolder	energiexp.nl/projecten/bathpolder/
Zonnepark Bavelse Berg	sweco.nl/portfolio/zonnecentrale-bavelse-berg/
Zonnepark Berkelweide	tpsolar.nl/berkelweide/
Zonnepark Billybird Park Drakenrijk	solarmagazine.nl/nieuws-zonne-energie/i35969/project-van-de-week-billybird-park-drakenrijk-neemt-drijvend-zonnepark-in-gebruik
Zonnepark Blakterweg	solarmagazine.nl/nieuws-zonne-energie/i34561/naga-solar-start-bouw-zonnepark-aan-blakterweg-in-venray
Zonnepark Blitsaerd	groenleven.nl/projecten/zonnepark-blitsaerd/
Zonnepark Bloesemlaan	tpsolar.nl/bloesemlaan/
Zonnepark Boeldershoek-West	solarmagazine.nl/nieuws-zonne-energie/i17137/twence-neemt-zonnepark-boeldershoek-west-in-gebruik-61-000-zonnepanelen-aangesloten
Zonnepark Bollendonk	statkraft.nl/nieuw/2023/opening-zonnepark/bollendonk/
Zonnepark Bontepolder	novar.nl/projecten/zonnepark-bontepolder-terneuzen/
Zonnepark Boxveenweg	chintsolar.com/nl/nieuws/eerste-zonnepaneel-gelegd-voor-zonnepark-boksveenweg
Zonnepark Branderwal	branderwal.nl
Zonnepark Budel	statkraft.nl/nieuws/2022/statkraft-opent-deuren-zonnepark-budel-nyrstar-op-open-energie-dag/
Zonnepark Buinerhord	chintsolar.com/nl/projectewn/zonnepark-buinerhorn/
Zonnepark Buinerveen	chintsolar.com/nl/projecten/zonnepark-buinerveen/
Zonnepark Coevorden	powerfield.nl/projecten/zonnepark-coevorden
Zonnepark Coevorderkanaal	winvast.nl/projecten/projecten-operationeel/zonnepark-coevorderkanaal/
Zonnepark Cothen	sunvest.nl/portfolio/zonnepark-cothen
Zonnepark Crayestein	hvcgroep.nl/ons-verhaal/onze-groene-projecten/zonnepark-crayestein-dordrecht
Zonnepark D'Aasdonken	energiexp.nl/projecten/daasdonken/
Zonnepark Daalkampen	dejongodoorn.nl/?page_id=1496

Zonnepark Daarle	solarproactive.nl/project/zonneakker-revelhorst/
Zonnepark Dassenberg	odera.nl/portfolio_page/zonneveld-dassenberg/
Zonnepark De Boomgaard - Wijdmeren	novar.nl/projecten/zonnepark-de-boomgaard-wijdmeren/
Zonnepark de Dijken	spectral.energy/nl/project/de-dijken-solar-power-meets-mobile-batteries
Zonnepark de Dogger	profinrg.nl/zonnepark-de-dogger-den-helder/
Zonnepark De Dreef	powerfield.nl/projecten/zonnepark-de-dreef/
Zonnepark De Elsmoat	profinrg.nl/profinrg-start-3-projecten-in-enschede/
Zonnepark de Kie	novar.nl/projecten/zonnepark-de-kie-franeker/
Zonnepark de Roodmolen	profinrg.nl/profinrg-start-3-projecten-in-enschede/
Zonnepark de Stegenhoek	lcenergy.nl/projecten/zonnepark-de-stegenhoek/
Zonnepark De Watering	novar.nl/projecten/zonnepark-de-watering-coevorden/
Zonnepark de Zwette	groenleven.nl/projecten/zonnepark-de-zwette/
Zonnepark Dorhout Mees	novar.nl/projecten/zonnepark-dorhout-mees-biddinghuizen/
Zonnepark Dorkwerd	novar.nl/projecten/zonthermiepark-dorkwerd-groningen/
Zonnepark Drouwenerzon	chintsolar.com/nl/projecten/zonnepark-drouwenerzon/
Zonnepark Duurkenakker	sunvest.nl/portfolio/zonnepark-duurkenakker/
Zonnepark EcoPark	langstraatzon.nl
Zonnepark Edelhertweg	linkedin.com/posts/circul8-energy_zonnepark-edelhertweg-zomer-2023-afgelopen-activity-7140621219921317888-IPgi/?originalSubdomain=nl
Zonnepark Eemscentrale	engie.nl/over-ons/moderne-centrales/eemscentrale
Zonnepark Eerbeek	brummenenergie.nl/project/zonneparken/zonnepark-eeerbeek/
Zonnepark Efteling	efteling.com/nl/blog/corporate/20230119-aanleg-zonneweide-met-12000-extra-zonnepanelen
Zonnepark Eimersweg	chintsolar.com/nl/projecten/zonnepark-eimersweg
Zonnepark Elhorst-Vloedbelt	twence.nl/projecten/zonnepark-elhorst-vloedbelt
Zonnepark Emmeloord	groenleven.nl/projecten/zonnepark-emmeloord/
Zonnepark Emmen	powerfield.nl/projecten/energieopslag-en-zonnepark-emmen/
Zonnepark Ewijk	gld.nl/nieuws/2411305/na-wat-hobbels-toch-start-zonnepark-ewijk
Zonnepark Farmsum - Delfzijl	novar.nl/projecten/zonnepark-vuilstort-farmsum/
Zonnepark Ficarystraat	tpsolar.nl/ficarystraat/
Zonnepark Fledderbosch	ectb.nl/fledderbosch/
Zonnepark Flevokust	solarmagazine.nl/nieuws-zonne-energie/i18841/engie-levert-zonnepark-flevokust-van-29-5-megawattpiek-op
Zonnepark Flierbelten	groenleven.nl/projecten/zonnepark-flierbelten/
Zonnepark Fluitenberg	powerfield.nl/projecten/zonnepark-fluitenberg/
Zonnepark Fort de Pol	novar.nl/projecten/zonnepark-fort-de-pol-zutphen/
Zonnepark Galecop	nmu.nl/nieuws/grootste-zonnepark-midden-nederland-opent-in-nieuwegein/
Zonnepark Garijp	groenleven.nl/projecten/zonnepark-garijp/
Zonnepark Geefsweer - Delfzijl	novar.nl/projecten/zonnepark-geefsweer-delfzijl/
Zonnepark Geesteren	ixzon.nl/geesteren/
Zonnepark Goor - Waterlanden	novar.nl/projecten/zonnepark-waterlanden-goor/
Zonnepark Groetpolder	zonneweidegroetpolder.nl
Zonnepark Groningen	groenleven.nl/projecten/zonnepark-groningen/

Zonnepark Haarstdiek	zonhub.com/project/haarstdiek/
Zonnepark Haarweg	lcenergy.nl/projecten/zonnepark-haarweg/
Zonnepark Harderwold	zonnepanelendelen.nl/project/harderwold/
Zonnepark Haulerwijk	groenleven.nl/projecten/zonneparken-ooststellingwerf/
Zonnepark Hazenwinkel	zonneparkhazenwinkel.nl
Zonnepark Heerenveen-Zuid	energieloketheerenveen.nl/initiatieven/zonnepark-heerenveen-zuid/
Zonnepark Heerhugowaard (De Vaandel)	zonnepanelenplanet.nl/zonneparken/noord-holland/zonnepark-heerhugowaard-de-vaandel/
Zonnepark Heldair I	ecorus.com/news/2021/eerste-graafwerk-voor-zonnepark-kooypunt/ / Wiki Lijst van installaties in Nederland voor vermoghe
Zonnepark Hemmen	solarmagazine.nl/nieuws-zonne-energie/i35413/izen-levert-zonnepark-hemmen-met-37-736-zonnepanelen-op
Zonnepark Hernen	installatie360.be/elektriciteit/extra-aandacht-
Zonnepark Het Broek Beuningen	profinrg.nl/zonnepark-het-broek-beuningen/
Zonnepark het Zonneward Zeewolde	profinrg.nl/zonnepark-het-zonneward-zeewolde/
Zonnepark Hijken	solarmagazine.nl/nieuws-zonne-energie/i90002/rechtbank-geeft-groen-licht-voor-zonnepark-hijken-in-midden-drenthe
Zonnepark Hoevelakensebeek	ixzon.nl/hoevelakensebeek/
Zonnepark Hoogveld-Zuid	tpsolar.nl/hoogveld-zuid/
Zonnepark Jumaheerd	groenleven.nl/projecten/zonnepark-jumaheerd/
Zonnepark Kampen	chintsolar.com/nl/projecten/zonnepark-kampen/
Zonnepark Kerkendijk	zonneparksomeren.nl/zonnepark-lungendonk/
Zonnepark Klarenbeek	de-a.nl/roject/zonneparken/zonnepark-klarenbeek/
Zonnepark Koningspleij	profinrg.nl/zonnepark-koningspleij-arnhem/
Zonnepark Kooypunt	vattenfall.nl/over-vattenfall/onze-energiebronnen/zonne-energie/zonneweide-kooypunt/
Zonnepark Koudekerke	zeeuwind.nl/zonneparken/koudekerke/
Zonnepark Kreekrakpolder	energiexp.nl/projecten/kreekrakpolder/
Zonnepark Laarberg	groendus.nl/klantverhalen/zonnepark-laarberg
Zonnepark Lange Runde	statkraft.nl/onze-energieprojecten/Nederland/zonnepark-lange-runde/
Zonnepark Leemdijk	energie.vanons.orh/bronnen/zonnepark-leemdijk/
Zonnepark Leveroydiedijk	soltronergy.eu/dt_portfolios/zonnepark-leveroydiedijk-weert/
Zonnepark Lievelede	sunvest.nl/portfolio/zonnepark-lievelede/
Zonnepark Lingewal	solarmagazine.nl/nieuws-zonne-energie/i18670/kronos-solar-start-bouw-zonnepark-lingewal-van-40-000-zonnepanelen
Zonnepark Louisegroeve	chemelot.nl/nieuws/start-bouw-grootste-zonnepark-van-limburg-op-chemelot-terrein
Zonnepark Lungendonk	tpsolar.nl/lungendonk/
Zonnepark Markelo	chintsolar.com/nl/projecten/zonnepark-markelo/
Zonnepark Marum	novar.nl/projecten/zonnepark-marum/
Zonnepark Masterveldweg	statkraft.nl/nieuws/2024/statkraft-opent-grootste-zonnepark-van-gelderland-in-winterswijk/
Zonnepark Medel	lcenergy.nl/projecten/zonnepark/medel/
Zonnepark Melissant	kuiipersinfra-strijen.nl/zonneparken-ooltgensplaat-en-melissant/
Zonnepark Middelburg	groenleven.nl/projecten/zonnepark-middelburg/
Zonnepark Middelsee	solarproactive.nl/project/zonnepark-middelsee/

Zonnepark Molenwaard	novar.nl/projecten/zonnepark-molenwaard-hoogezand/
Zonnepark Mosselbanken	solarmagazine.nl/nieuws-zonne-energie/i26817/zonnepark-mosselbanken-geopend-zeelands-grootste-telt-111-114-zonnepanelen
Zonnepark Munsterjansdijk	novar.nl/projecten/zonnepark-munsterjansdijk-eibergen/
Zonnepark Nieuweweg	zonhub.nl/project/nieuweweg/
Zonnepark Nijnselseweg	zonnepanelendelen.nl/project/nijnselseweg
Zonnepark Noordmanshoek	noordmanshoek.nl/zonnepark/
Zonnepark Noordscheschut	novar.nl/projecten/zonnepark-noordscheschut-hoogeveen/
Zonnepark Ommelandervijk	chintsolar.com/nl/projecten/zonnepark-ommelandervijk/
Zonnepark Omweg (Leeuwarderstraatweg)	energieloketheerenveen.nl/initiatieven/zonnepark-omweg-leeuwarderstraatweg/
Zonnepark Ooltgensplaat	kuipersinfra-strijen.nl/zonneparken-ooltgensplaat-en-melissant/
Zonnepark Oosterlaak	sunvest.nl/portfolio/zonnepark-oosterlaak/
Zonnepark Oosterweilanden	twenterand.nl/zonnepark-oosterweilanden
Zonnepark Oranjepoort Emmen	groenleven.nl/projecten/zonnepark-oranjepoort-emmen/
Zonnepark Oudendoorn	zonneparkoudendoorn.nl
Zonnepark Overbetuwe	profinrg.nl/zonnepark-overbetuwe/
Zonnepark Pottendijk	solarmagazine.nl/nieuws-zonne-energie/i34079/zonnepanelen-energiepark-pottendijk-van-shell-starten-stroomproductie
Zonnepark Rilland	solarmagazine.nl/nieuws-zonne-energie/i18898/zonnepark-rilland-geopend-europa-s-grootste-met-bifaciale-zonnepanelen
Zonnepark Roodehaan	novar.nl/projecten/zonnepark-roodehaan-groningen/
Zonnepark Sas van Gent Zuid	shell.nl/over-ons/nieuws/nieuwsberichten-2022/Shell-en-Terneuzen-openen-zonnepark-Sas-van-Gent-Zuid.html
Zonnepark Schaapsdijk	solarmagazine.nl/nieuws-zonne-energie-
Zonnepark Schalkwijk	groenleven.nl/projecten/zonnepark-schalkwijk/
Zonnepark Schoenaker	tpsolar.nl/schoenaker/
Zonnepark Sinnegreide - Achtkarspelen	novar.nl/projecten/zonnepark-achtkarspelen/
Zonnepark Solvay	greenchoiceintegrated.nl/projecten/solvay/
Zonnepark Stadskanaal	solarmagazine.nl/First Solar levert 20 megawattpiek zonnepanelen uit, bouw zonnepark Stadskanaal gestart
Zonnepark Stadskanaal 1	solarmagazine.nl/nieuws-zonne-energie/i21195/bouw-zonnepark-stadskanaal-van-start-100-megawattpiek-zonnepanelen-op-komst
Zonnepark Sunspace	sunspace.nl
Zonnepark Tholen	eneco.nl/over-ons/wat-we-doen/duurzame-bronnen/zon/zonnepark-Tholen/
Zonnepark Tinte / Voorne aan Zee	voorneputtenenergie.nl/zonnepark-tinte
Zonnepark Transberg	tpsolar.nl/transberg/
Zonnepark Tripkouw	hvcgroep.nl/ons-verhaal/onze-groene-projecten/zonneweide-tripkouw
Zonnepark Valgenweg	sun-projects.nl/nl/nieuws/opening-zonnepark-valgenweg
Zonnepark Van der Duijnsplander	energiexp.nl/projecten/van-der-duijnsplander/
Zonnepark Veenweg	lcenergy.nl/projecten/zonnepark-veenweg/
Zonnepark Vegelinsoord	greenchoiceintegrated.nl/projecten/klimaatfonds-vegelinsoord
Zonnepark Vlagtwedde I	powerfield.nl/projecten/zonnepark-vlagtwedde/
Zonnepark Vleddermond Stadskanaal	solarmagazine.nl/ novar.nl/projecten/zonnepark-vleddermond-stadskanaal/

Zonnepark Vloelvelden Hollandia - Nieuw-Buinen	novar.nl/projecten/zonnepark-vloelvelden-hollandia/
Zonnepark W.A. Scholten	novar.nl/projecten/zonnepark-w-a-scholten-foxhol/
Zonnepark Wehl	doetinchem.nl/zonneparken-en-windparken
Zonnepark Wenumseveld	statkraft.nl
Zonnepark Werkhoven	lcenergy.nl/projecten/zonnepark-werkhoven/
Zonnepark Westdorp	lightsourcebp.com/app/uploads/2022/12/Community-Engagement-Pack-Zonnepark-Westdorp_LR-ID-253195-2.pdf
Zonnepark Wieldrechtse Zeedijk	novar.nl/projecten/zonnepark-wieldrechtse-zeedijk-dordrecht/
Zonnepark Wijkerbroek Oost	lcenergy.nl/projecten/zonneveld-wijkerbroek-oost/
Zonnepark Wijkerbroek West	lcenergy.nl/projecten/zonneveld-wijkerbroek-west/
Zonnepark Wilbertoord	statkraft.nl/nieuw/2024/statkraft-opent-zonnepark-wilbertoord-in-gemeente-land-van-cuijk/
Zonnepark Wolvega	powerfield.nl/projecten/zonnepark-wolvega/
Zonnepark Zeeland Solar	zeelandrefinery.nl/zeeland-refinery-start-bouw-zonnepark-zeeland-solar/
Zonnepark Zierikzee	schouwen-duiveland.nl/duurzaamheid-en-milieu/spoor-1-energietransitie/duurzame-energieopwekking/projecten-zonne-energie
Zonnepark Zonnestroom	sunvest.nl/portfolio/zonnepark-revelhorst/
Zonnepark Zuidbroek	profinrg.nl/zonnepark-zuidbroek/
Zonnepark Zuidgeest	solarprovidergroup.nl/projecten/zuidgeest-(bergen-op-zoom)
Zonnepark Zuidplas	zonnepark-zuidplas.nl
Zonnepark Zuyderzon	hvcgroep.nl/ons-verhaal/onze-groene-projecten/zonnepark-zuyderzon-almere
Zonneveld A1 Eemnes	steunpuntcultureelerfgoednh.nl/zonneveld-eemnes-als-een-kamer-in-het-landschap/
Zonneveld Best	zonnestroomnederland.nl/projecten/zonneveld-best/
Zonneveld Meander	ad.nl/amersfoort/plan-voor-zonneveld-naast-meander-medisch-centrum~a7320291
Zonneweide Echteld Spoorstraat	vattenfall.nl/over-vattenfall/onze-energiebronnen/zonne-energie/zonneweide-spoorstraat/
Zonneweide Haghorst	meewind.nl/projecten/zonnepark-haghorst/

Appendix B: Land use classes

LGN 7	LGN2024	Generalized classes
agrarisch gras	agrarisch gras	1 - Agricultural area
mais	mais	1 - Agricultural area
aardappelen	aardappelen	1 - Agricultural area
bieten	bieten	1 - Agricultural area
granen	granen	1 - Agricultural area
overige landbouwgewassen	overige landbouwgewassen	1 - Agricultural area
glastuinbouw	glastuinbouw	2 - Greenhouse horticulture
boomgaarden	boomgaarden	3 - Orchards
bloembollen	bloembollen	1 - Agricultural area
loofbos	loofbos	4 - Forest
naaldbos	naaldbos	4 - Forest
zoet water	zoet water	5 - Water
zout water	zout water	5 - Water
bebouwing in primair bebouwd gebied	bebouwing in primair bebouwd gebied	6 - Urban area
bebouwing in secundair bebouwd gebied	bebouwing in secundair bebouwd gebied	6 - Urban area
bos in primair bebouwd gebied	bos in primair bebouwd gebied	6 - Urban area
bos in secundair bebouwd gebied	bos in secundair bebouwd gebied	6 - Urban area
gras in primair bebouwd gebied	gras in primair bebouwd gebied	6 - Urban area
kale grond in bebouwd gebied	-	6 - Urban area
wegen en spoorwegen	-	7 - Infrastructure
bebouwing in buitengebied	bebouwing in buitengebied	1 - Agricultural area
-	overig grondgebruik in buitengebied	1 - Agricultural area
gras in secundair bebouwd gebied	gras in secundair bebouwd gebied	6 - Urban area
-	zonneparken	9 - Solar parks
kwelders	kwelders	8 - Nature
open zand in kustgebied	open zand in kustgebied	8 - Nature
duinen met lage vegetatie	duinen met lage vegetatie	8 - Nature
duinen met hoge vegetatie	duinen met hoge vegetatie	8 - Nature
duinheide	duinheide	8 - Nature
open stuifzand en/of rivierzand	open stuifzand en/of rivierzand	8 - Nature
heide	heide	8 - Nature
matig vergraste heide	matig vergraste heide	8 - Nature
sterk vergraste heide	sterk vergraste heide	8 - Nature
hoogveen	hoogveen	8 - Nature
bos in hoogveengebied	bos in hoogveengebied	8 - Nature
overige moeras vegetatie	overige moeras vegetatie	8 - Nature
rietvegetatie	rietvegetatie	8 - Nature
bos in moerasgebied	bos in moerasgebied	8 - Nature

natuurlijk beheerde agrarische graslanden	natuurlijk beheerde agrarische graslanden	1 - Agricultural area
-	gras in het kustgebied	8 - Nature
-	overig gras	8 - Nature
boomkwekerijen	boomkwekerijen	1 - Agricultural area
fruitkwekerijen	fruitkwekerijen	3 - Orchards
-	hoofdinfrastructuur en spoorbaanlichamen	7 - Infrastructure
-	halfverharde wegen, infrastructuur langzaam verkeer en overige infrastructuur	7 - Infrastructure
-	smalle wegen	7 - Infrastructure
-	struikvegetatie in hoogveengebied (laag)	8 - Nature
-	struikvegetatie in moerasgebied (laag)	8 - Nature
-	overige struikvegetatie (laag)	8 - Nature
-	struikvegetatie in hoogveengebied (hoog)	8 - Nature
-	struikvegetatie in moerasgebied (hoog)	8 - Nature
-	overige struikvegetatie (hoog)	8 - Nature

Appendix C: Scene dates for the Landsat composites used for LST analysis

2015	2025
24-05-2015	01-05-2025
07-06-2015	10-05-2025
02-07-2015	12-05-2025
04-07-2015	19-05-2025
11-07-2015	11-06-2025
18-07-2015	13-06-2025
03-08-2015	18-06-2025
05-08-2015	20-05-2025
21-08-2015	13-07-2025
	07-08-2025
	14-08-2025

Appendix D: National transition matrix

Table F1: Land-use transitions on the national level between 2012 (LGN7) and 2024 (LGN2024). Rows represent initial land use, and columns represent land use in 2024. Percentages refer to the share of each 2012 land-use class; values in *italic* below them indicate area in hectares (rounded to whole hectares). Diagonal cells indicate land-use persistence or “unchanged” (written in *green*).

	LGN2024														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
LGN7	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	70.12% 888940	5.83% 73866	5.01% 63546	0.07% 837	0.06% 698	0.42% 5378	4.92% 62417	4.58% 58065	2.81% 35578	2.53% 32085	0.14 %	3.06% 38821	0.23% 2901	0.12% 1584	0.11% 1362
2	37.73% 103112	29.76% 81323	22.93% 62673	0.03% 86	0.02% 46	0.89% 2443	1.14% 3111	2.86% 7823	1.1% 3003	1.82% 4976	0.08% 214	0.43% 1185	0.66% 1811	0.29% 784	0.25% 679
3	13.2% 73536	7.81% 43484	70.17% 390817	0.1% 582	0.01% 48	1.92% 10679	0.71% 3931	2.46% 13695	0.69% 3869	1.44% 8037	0.05% 305	0.33% 1852	0.34% 1880	0.54% 3029	0.22% 1250
4	4.5% 582	0.46% 59	0.79% 103	84.46% 10927	0.01% 2	0.24% 31	0.04% 5	0.92% 119	3.48% 450	1.16% 150	0.48% 63	2.62% 339	0.52% 67	0.3% 38	0.03% 4
5	29.05% 711	0.97% 24	1.92% 47	0.03% 1	45.61% 1116	0.07% 2	3.69% 90	3.04% 75	4.08% 100	3.63% 89	0.14% 4	3.05% 75	0.56% 14	4.14% 101	- 0
6	19.98% 5610	6.03% 1692	29.17% 8188	0.1% 29	0.01% 2	36.79% 10327	0.22% 61	2.64% 742	0.5% 141	1.56% 438	0.11% 32	0.51% 144	1.88% 527	0.37% 103	0.14% 39
7	1.95% 9670	0.16% 788	0.29% 1419	- 4	0.01% 34	0.01% 26	86.29% 428015	3.43% 17010	3.03% 15015	4.45% 22050	0.01% 63	0.32% 1570	0.04% 214	0.01% 34	0.02% 90
8	0.33% 2666	0.01% 43	0.01% 77	- 19	- 3	- 12	1.03% 8335	97.47% 789561	0.23% 1885	0.84% 6790	- 17	0.05% 426	- 15	- 4	0.03% 214
9	2.38% 11589	0.07% 339	0.12% 568	0.04% 178	0.03% 167	0.01% 48	1.37% 6691	2.05% 9987	83.1% 405269	9.59% 46776	0.16% 791	0.9% 4378	0.03% 135	0.01% 61	0.15% 718
10	17.92% 15076	0.26% 223	0.49% 415	- 3	0.02% 16	0.02% 16	3.83% 3221	5.5% 4630	19.6% 16489	51.08% 42971	0.06 47	1.12% 942	0.03% 29	0.04% 33	0.02% 21
11	10.52% 9329	0.73% 643	0.83% 737	0.69% 615	0.15% 132	0.08% 72	2.08% 1843	2.56% 2268	17.75% 15730	2.06 1825	22.9% 20301	39.25% 34796	0.2% 179	0.18% 156	0.02% 16
13	16.42% 3814	8.46% 1963	16.74% 3883	0.38% 88	0.07% 15	2.33% 541	1.41% 326	3.85% 894	1.77% 411	2.67% 619	0.19% 44	1.96% 454	41.4% 9603	2.19% 508	0.13% 29
14	8.62% 1990	3.14% 724	10.81% 2495	0.11% 26	0.5% 116	0.55% 127	0.46% 105	3.25% 750	1.01% 233	2.19% 505	0.12% 28	0.79% 183	0.99% 229	67.26% 15529	0.2% 47

Note: The two last columns “Other land use in rural areas” and “Solar parks” were added in LGN 2024. Values with “-” (no percentage) indicate that < 0.01 transitioned.

Table F2: *Accompanying land use number table*

Number	Name
1	Agricultural grass
2	Corn
3	Arable/horticultural crop
4	Greenhouse horticulture
5	Orchards
6	Flower bulbs
7	Forest, trees and nature
8	Water
9	Urban area
10	Infrastructure
11	Buildings in rural areas
12	Other land use in rural areas
13	Tree nurseries
14	Fruit nurseries
15	Solar parks

Appendix G: Dominant land-use transitions in buffer areas around solar parks (2012–2024)

Note: Each buffer overlaps multiple land-use classes in 2012. The dominant land use is defined as the class covering the largest share of the buffer area. This table includes only the 16 parks where the dominant land use changed in either the 500 m or 2 km buffer. For the remaining 190 parks, the dominant land use remained unchanged in both buffers.

Solar park	Province	Prior land use	500m transition	2km transition
Zonnecarport TT Circuit Assen	Drenthe	Infrastructure	Urban area -> Nature	Agricultural area -> Nature
Zonnepark Harderwold	Flevoland	Urban area	Urban area -> Nature	Unchanged (Water)
Zonnepark Blitsaerd	Fryslân	Agricultural area	Agricultural area -> Urban area	Unchanged (Agricultural area)
Drijvend Zonnepark de Uivermeertjes	Gelderland	Water	Agricultural area -> Water	Unchanged (Agricultural area)
Zonnepark Apeldoorn	Gelderland	Agricultural area	Agricultural area -> Urban area	Unchanged (Agricultural area)
Zonnepark Eerbeek	Gelderland	Urban area	Forest -> Urban area	Unchanged (Agricultural area)
Zonnepark Fort de Pol	Gelderland	Urban area	Urban area -> Agricultural area	Unchanged (Agricultural area)
Zonnepark Zonnestroom	Gelderland	Agricultural area	Agricultural area -> Urban area	Unchanged (Agricultural area)
Zonnepark Groningen	Groningen	Urban area	Agricultural area -> Urban area	Unchanged (Urban area)
Zonnepark Louisegroeve	Limburg	Urban area	Unchanged (Urban area)	Urban area -> Agricultural area
Zonnepark Bavelse Berg	Noord- Brabant	Agricultural area	Unchanged (Agricultural area)	Agricultural area -> Urban area
Zonnepark de Boomgaard – Wijdmeren	Noord- Holland	Urban area	Nature -> Agricultural area	Unchanged (Nature)
Zonnepark Aadijk	Overijssel	Agricultural area	Unchanged (Agricultural area)	Agricultural area -> Urban area
Zonnepark Oosterweilanden	Overijssel	Agricultural area	Agricultural area -> Urban area	Unchanged (Agricultural area)
Zonnepark Cothen	Utrecht	Orchards	Agricultural area -> Orchards	Unchanged (Agricultural area)

Zonnepark Zeeland Solar	Zeeland	Agricultural area	Unchanged (Urban area)	Agricultural area -> Urban area
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